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INTRODUCTION

Did you buy this book because you wanted to know about Formula Racing? No problem; we will give you a short introduction to what Formula Racing is.

Formula Racing is a motorsport of thin, light cars with open cockpits for drivers.

Every year there is an official F1 Grand Prix hosted by the FIA (Federation Internationale d'Automobile, which you can read more about in Strategy and Profits Garage).

There are 11 teams as of the 2026 season.

The most famous teams are Ferrari, Mercedes, Williams Racing, Red Bull, etc.

Each team has 2 cars, which they build by themselves, and 2 drivers.

The current famous drivers are Max Verstappen, Andrea Kimi Antonelli, Lando Norris and more.

Each team gets points with which they can win.

Cars are built with the most high-tech materials, like Aerofoil and Titanium. To find out more, turn the page…

FORMULA RACING CAMP

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CHAPTER 01

FUTURE MECHANICS

Hello future champions, I'm Jaskirat- a pro, prestigious and prime F1 driver of the Neon Raptor Vortex GT-X with my pet, Igneous the lightning, shape-shifting iguana, so he can turn into almost everything, including different types of Iguana!- Remember that now! And welcome to the first chapter of Formula Racing Camp. Here, we'll talk about aerodynamics in F1, wings, and how they affect awesome races!

Aerodynamics

First of all, what is aerodynamics - the air doing gymnastics? Umm, no. Aerodynamics is the study of how air moves around objects and how that movement affects them. In simple words:
It explains how air pushes, pulls, or flows around things that move, like cars,
bikes, aeroplanes, or even a football. Aerodynamics is how air decides to be annoying.

So we make cars, bikes, and planes **smooth and pointy** so air goes:
"Oh okay, I'll just slide past!"

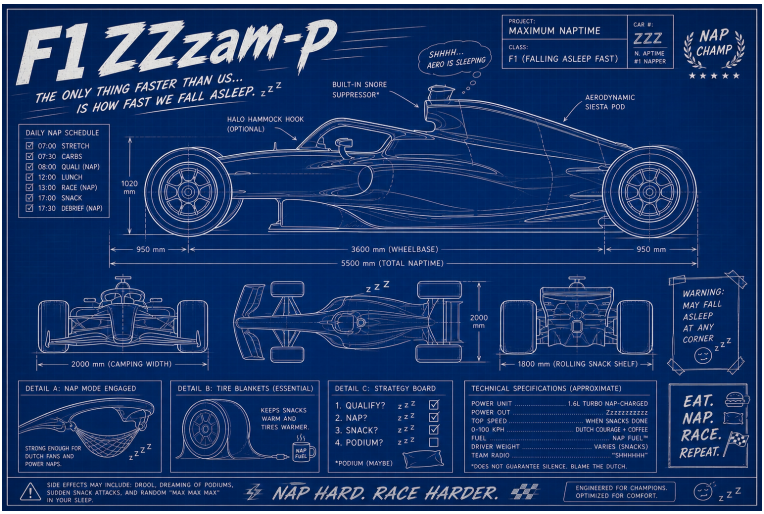
If air hits you hard → you go slow.
If air slips nicely around you → Speed!

It's the car's invisible superpower, controlling air to make it faster, stickier in corners, and more stable at insane speeds. The wings push the car down, creating downforce—so strong that, in theory, an F1 car could drive upside down in a tunnel! Without aerodynamics, cars would be like kites at 300 km/h, sliding and struggling to stay on track. Here's a joke: Why don't F1 cars ever get lost? Because they always follow the downforce!

If aerodynamics didn't exist, F1 wouldn't really exist & so wouldn't this super cool F1 book (That is BAD!!). Here's why: Formula 1 cars rely on aerodynamics to generate downforce, which pushes the car onto the track and gives it grip.

Without it, cars would slide around in corners, struggle to stay stable at high speeds, and could even lift off slightly at over 300 km/h. Races would be

slower, far less precise, and much more dangerous—so the sport would be completely different, more like high-speed racing than the extreme engineering spectacle we see today. **Joke Time:** Scientists & coaches are still investigating why this wingless F1 car makes drivers instantly sleepy. Some blame the smooth aerodynamics, some blame the ultra-comfy cockpit… but after 3 laps, every test driver started snoring. That's why we proudly call it the **F1 ZZZAM-P!** — the fastest machine from 0 to record time!!



Wings

The wings on an F1 car are used to control the air so the car can go faster without sliding off the track. When an F1 car moves, air flows over and under its wings. The wings are shaped in such a way that air pushes the car down toward the road instead of lifting it. This pushing-down force is called downforce. Downforce makes the tyres grip the road better, especially when the car is turning at very high speeds. Without wings, an F1 car would skid, slide, or even lose control in corners. So even though the wings look cool, their real job is to help the car stay stable, fast, and safe. **Oh great, Igneous made a toy line figure by shape-shifting into one! But it shows you where the different types of wings an F1 car has!**

Why F1 wings are important:

• **More grip** – The car sticks to the road like a magnet

• **Better cornering** – Cars can turn fast without slipping

• **Higher speed** – Drivers can go faster safely

• **More control** – The car stays balanced during braking and turning

So, F1 wings make the car hug the road, not fly in the air!

The History of Aerodynamics in F1:

1. 1950s - 1960s: Cars had no wings, just sleek bodies to reduce drag. Drivers relied on engine power and skill. And Igneous hates it!

2. Late 1960s: First wings appeared to push cars down for more grip in corners. Early high wings were unstable and risky.

3. 1970s: Ground effect introduced—shaping the car's underside to suck it onto the track for faster cornering.

4. 1980s–1990s: Wings and bodywork became more refined; wind tunnels and computers used to optimize aerodynamics.

5. 2000s: Cars had complex wings and diffusers; airflow management became key for speed and stability.

6. 2022s–Today: Modern F1 cars balance downforce and drag, with ground-effect floors returning to allow closer racing and more overtakes.

Today's Formula 1 cars are not just machines—they are masterpieces of airflow engineering. Every curve, wing, and surface is designed to control invisible streams of air with extreme precision. Teams use advanced simulations and powerful computers to test thousands of designs before a car even hits the track. The return of ground effect has transformed racing, letting drivers chase each other wheel-to-wheel at incredible speeds. In this era, victory doesn't just belong to the fastest engine—it belongs to the team that can command the air itself!

The Cost of an F1 Car's Wings

F1 cars' wings are very expensive because they are made from carbon fibre, designed in wind tunnels, and tested again and again.

• Front wing: roughly $150,000–$300,000

• Rear wing: roughly $100,000–$250,000

• With development and testing, a full wing design can cost millions over a season.

That's because even tiny shape changes can affect speed, grip, and lap time.

My car's wing, which was built in the UK and upgraded in Maranello, Italy, has my name built into the carbon itself, and when the car powers up, electric & flame energy ignite along the bottom of the wing—for looks, speed, and a warning to thieves. It has advanced aerodynamics – still produces strong downforce while hiding the tech inside. The shock system is for super-fast speed, and with Igneous, it powers up a lot more!

Wheels of F1 Cars (Slick tyres)

Ah yes! I almost forgot about the **smooth, cartoon-like wheels of F1 cars**—also called **slick tyres**—one of the most recognisable features on track, and a **huge reason** why F1 cars are insanely fast.

Unlike regular road tyres, F1 tyres have no tread patterns, which makes them perfectly smooth. You might wonder, "Won't that make them slippery?" Actually… It's the opposite! Good! I thought Igneous would think cartoons are more important than them!

Slick tyres are designed to maximise contact with the track, giving the car the most grip possible. More rubber touching the asphalt = more friction = faster cornering and braking.

Which means drivers can push harder through turns without losing control.

F1 tyres are engineered to handle crazy speeds and temperatures—up to 100°C or more! That's hotter than a frying pan! (And yes… no idea how frying pans entered the chat either).

Now imagine driving an F1 car on a rainy day with slick tyres. Do you think the car would grip the road well, or slide everywhere like it's on soap?
(Hint: slick tires love the dry tracks, not puddles !)

That's why F1 teams choose different tyres for different conditions —soft, medium, hard, and wet—to balance speed and durability, and drivers are trained to handle almost all types.

So those smooth, "cartoon" tyres aren't just for looks — they're high-tech performance machines. Okay… maybe they are also for looks, but mostly it's about performance!

If F1 Tires Were Normal Tires

But if the cool, unique F1 tyres were replaced with normal road car tyres, the car would be almost impossible to drive!!

See, F1 cars are super light, extremely fast, and have huge wings that push the car hard onto the road. Normal tyres are not made to handle that much speed, force, and heat. As soon as the car tried to go fast or turn, the tyres would lose grip, slide, or even burst. The car wouldn't be able to use its aerodynamics properly because the tyres wouldn't "hold on" to the road.

This is what would go wrong:

- **Very little grip** – normal tyres would slip and slide
- **Overheating** – they'd melt or wear out very fast
- **No fast cornering** – the car would spin or crash
- **Poor braking** – stopping would take much longer
- **Possible tyre failure** – at F1 speeds, they could burst!

The Engine (Heart of an F1 Car)

Alright, campers… We've already explored the wings that push the car down and the tyres that grip the track. But now, it's time to uncover the real beast hiding inside the machine… the engine — or as F1 calls it, the power unit.

The power unit provides the energy that helps the car accelerate and perform on track. In the simplest way possible, it's the heart of the car.

- **No engine = no speed**
- **No speed = no race**

And no race? Yeah… that's a disaster!

But F1 Engines Are NOT Normal

Here's where things get crazy. F1 engines are nothing like the ones in normal road cars. These machines use a super advanced hybrid power unit, meaning they use both fuel and electricity together.

That combination creates insane power that pushes the car to extreme limits.

How Powerful Is It?

So how powerful are we talking? Modern F1 power units produce around 1000 horsepower, depending on the car and conditions. Not actual horses—but the force of multiple cars combined into one machine.

- **Speeds over 350 km/h**

That's what makes an F1 car feel like a rocket on wheels.

Special Features

What makes it even cooler is how smart this system is.

F1 cars use an Energy Recovery System (ERS), which captures energy during braking and from exhaust heat, then reuses it to provide extra power-like a hidden boost button.

They also use a turbocharger, which compresses more air into the engine so more fuel can burn efficiently and produce more power.

Why the Engine is Important

The engine is one of the most important parts of the car.

- **Raw speed**
- **Acceleration**
- **Overtaking power**

It works together with aerodynamics to push performance to the limit.

Cost of an F1 Engine

Now for the shocking part… the price. An F1 power unit costs around $10–15 million (₹80–120 crore).

That's because it's not just an engine—it's a hybrid masterpiece built with cutting-edge technology, advanced turbo systems, energy recovery tech, and materials that can handle extreme conditions. Even the smallest upgrade can change the outcome of a race.

Famous Racers & Their Unique Engines

Over the years, legendary drivers have used some of the most powerful and iconic engines in F1 history.

Lewis Hamilton drove cars powered by Mercedes hybrid engines, known for their incredible efficiency and smooth power delivery. These engines helped him dominate the turbo-hybrid era and win multiple championships.

Michael Schumacher became a legend with Ferrari V10 engines in the early 2000s. These engines were famous for their screaming sound and raw power, helping Ferrari dominate F1 during that time.

Ayrton Senna used the iconic Honda turbo engines in the late 1980s. These engines were extremely powerful and aggressive, perfectly matching his
fearless driving style.

Max Verstappen currently uses Red Bull Ford Powertrains (Honda-based hybrid engines). These engines are known for their explosive acceleration and strong performance, helping him dominate modern F1 races.

F1 engines aren't just machines…
They're legend-makers!

Different engines + different drivers = different eras of domination

My Engine

The Neon Raptor Vortex GT-X doesn't just have AN engine…

It has a **Quantum Storm Hyper-Drive Core!**

This isn't some normal hybrid system.
This is a fusion of **lightning, fire, and pure racing energy**.

What Powers It?

• **Lightning Core Reactor** – Generates electric bursts faster than a thunderstrike!

• **Igneous Flame Sync** – My iguana literally powers the engine with living fire energy while standing on the wing!

• **Turbo Cyclone Compressors** – Spin air into a mini storm for insane acceleration!

What Happens When It Starts…

When I press the ignition:

The engine doesn't just "start"...

IT AWAKENS

• The car glows with electric energy

• Flames ripple across the rear vents

• The sound? Not a normal roar…

It sounds like a **thunderstorm ripping through a volcano!** (Weird description but okay…)

Performance Mode

When I activate full power:

- Energy surges through the chassis
- Flames boost acceleration
- Air bends around the car

The car doesn't just go fast…
It TELEPORTS forward like a lightning strike!

JustJoking-O-Rama Onslaught™!

Ok, campers… It's joke time! Igneous & I made some windy, rolly & growly jokes here, perfect for this chapter. Get ready for your first JustJoking-O-Rama.

Onslaught™!

• **Why did the aerodynamicist get kicked out of the library?** He kept trying to adjust the "drag" on the floor with the books!

• **How do F1 engineers stay cool?** They have a lot of fans... and they're all in the wind tunnel!

• **Why was the track so stuffy?** It's not 'cause the crowd left, well, maybe, but it's because there are messy tyres, rusty engines & everything lying on the tracks!

Igneous, don't spread hot air everywhere!

• **Why did the tyre cross the road?** It didn't. Why do you ask? He didn't want to!

• **Why did the tyre go to therapy?** Because it couldn't handle the pressure!

• **What does the last F1 driver of the race like to eat?** The dust of the others' tyres!

• **What's an F1 car's tyres' favourite type of music?** Rock n' Roll!

• **What did the tyre say after the race?** He said he was very tired!

But Igneous is 100000+ times more tired as a tire than a normal tire!

- **Why are F1 tires so bad at keeping secrets?** Because they always leak information through telemetry!
- **Why was the F1 engine so confident?**
  Because it had a lot of horsepower behind it!
- **What did the engine say before the race started?**
  "Let's make this a rev-olution!"
- **Why don't F1 engines ever get tired?**
  Because they're always revved up and ready to go!

Chapter 1 Quiz!

Aerodynamics, Wings, Tires & Engines

Part A – Multiple Choice (1 mark each)

1. What is aerodynamics?

A) The study of engines
B) The study of how air moves around objects
C) The study of tyres
D) The study of fuel

2. Downforce helps an F1 car...

A) Fly into the air
B) Stick to the road
C) Use less fuel
D) Change tyres

3. Which part mainly creates downforce?

A) Steering wheel
B) Wings
C) Mirrors
D) Seat

4. Why are F1 tyres smooth?

A) They are cheaper
B) They create more grip on dry tracks
C) They look better
D) They reduce weight

5. Which tyre is best for heavy rain?

A) Slick
B) Soft
C) Wet tyre
D) Medium

6. Modern F1 power units use...

A) Steam
B) Petrol only
C) Fuel and electricity
D) Solar panels

7. ERS stands for...

A) Engine Racing Speed
B) Energy Recovery System
C) Engine Rotation System
D) Electric Racing Setup

8. Approximately how much horsepower does a modern F1 car produce?

A) 300 hp
B) 500 hp
C) 700 hp
D) Around 1000 hp

9. Which material are F1 wings mainly made from?

A) Steel
B) Aluminium
C) Carbon fibre
D) Plastic

10. Ground effect mainly works by...

A) Cooling the engine

B) Sucking the car towards the track

C) Reducing fuel use

D) Increasing tyre size

Part B – True or False

11. Aerodynamics only matters for aeroplanes.

12. More downforce usually gives better cornering.

13. Slick tyres work best in heavy rain.

14. Modern F1 cars use hybrid power units.

15. Without wings, F1 cars would be harder to control.

16. Engines and aerodynamics work together.

17. Formula 1 began using wings in the late 1960s.

18. Carbon fibre is heavier than steel.

19. Wind tunnels help engineers design wings.

20. An F1 engine costs millions of dollars.

Part C – Fill in the Blanks

21. _____ is the force that pushes the car onto the track.

22. Smooth F1 tires are called _____ tires.

23. F1 wings are mainly made from _____ fibre.

24. The _____ stores and reuses braking energy.

25. Modern F1 cars can reach speeds above _____ km/h.

26. The engine is often called the _____ of the car.

27. Ground effect helps increase _____.

28. More rubber touching the road creates more _____.

29. Air flowing around the car affects its _____.

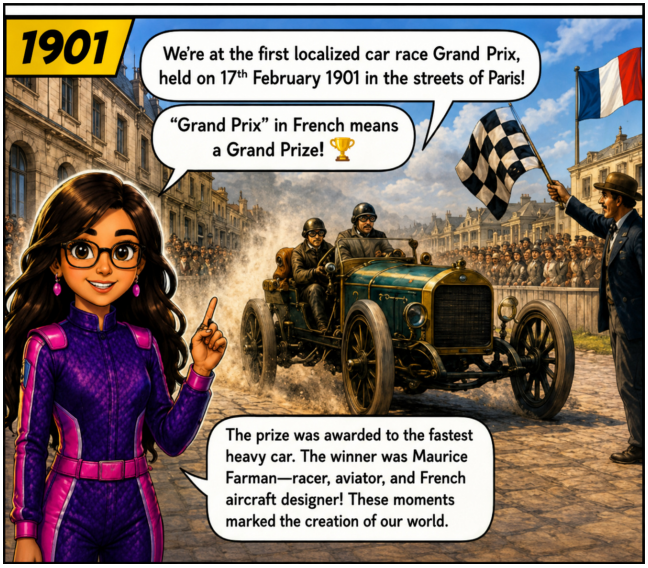
30. Formula 1 first started in _____.

CHAPTER 02

TIME TRAVELLERS

Hi Campers! Welcome to the time traveller lab. Formula Racing Camp's lab of the past. The lab has a very cool machine, the Past-O-Meter. The machine is a time traveller. Isn't that cool? Today, this machine is going to help us to explore the past of Formula. Today we will explore the evolution of the car. So take your seats in the Past-O-Meter, so we can go to the start of F1 racing. **Past-O-Meter says, "Transporting to the 1901 Grand Prix in Pau."**

1901



So here we are, at the first localised car race Grand Prix held before F1 had begun on 17th February 1901 in the streets of Pau. **Did you know the Grand Prix in French means a Grand Prize?** In this race, the prize was awarded to the fastest heavy car. The rumbles of these cars on these very streets you are seeing are where our racing world was first created. As you can see, the flag was waved, and these early petrol vehicles rumbled their engines. These moments are what marked the creation of our world. **Did you know the Grand Prix's winner was Maurice Farman!** He was not only a racer but an aviator, French aircraft designer and manufacturer. The Grand Prix was a significant part of racing history.

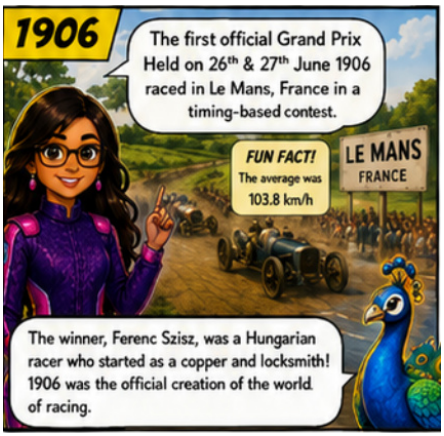
1904

Now let us move forward in time. Past -O -meter voice, "Teleporting you to 20th June 1904 in France." Now come on, drivers ride into the meeting that laid the foundation for the creation of the FédérationInternationale de l' Automobile (FIA). Their headquarters are in Paris, France. This marked the start of efforts on safety and management of the Car Racing body sport. As we all know, safety is crucial in our field because of the risk of accidents at the speeds we go. The world of racing was safer than it was before.

1906

Now, campers, let us move to 1906 when the first official Grand Prix was held, since the one held in 1901 was a localised unofficial race.

Did you know that this Grand Prix was held around Le Mans, France, on a triangular circuit (**Fun Fact: It measured 103.8 km)?** The Grand Prix was held on 26th and 27th June of 1906. Did you know that the winner, Ferenc Szisz, is a Hungarian racer who started as a copper and locksmith (The only copper he used later in life was in his cars!), but then the amazing power of automobiles fascinated him to be an automobile engineer, which led him to the world of racing!

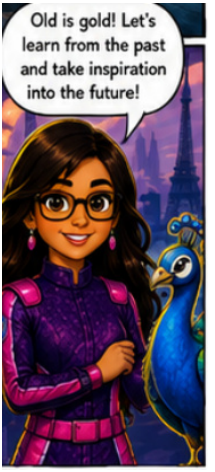


Parts Of The Early 1900s Cars:

This is to explore the parts of the old car. As they say, old is gold; let's inspire ourselves with this and have fun. So let us ride in.

We will explore this using the Mercedes 35hp **(Fun Fact: Known as the first ever modern car)**

1. Engine - The heart of the car had 4 Cylinders In Line Engine This means the engine is not only made of metal tubes, which causes the car to vibrate heavily and then pause, but it is made up of four metal tubes, which causes the vibration to go in a rhythm, making the car go faster. **Did you know that this framework is present as the basic structure of the present-day cars?**



Parts of the Engine :

• **4 cylinders:** allowed the car to reach speeds like 80km/hr which was super fast for that time. This allowed a 5.9 L displacement.

• **Crankcase:** A long metal bar housing around the Crankshaft.

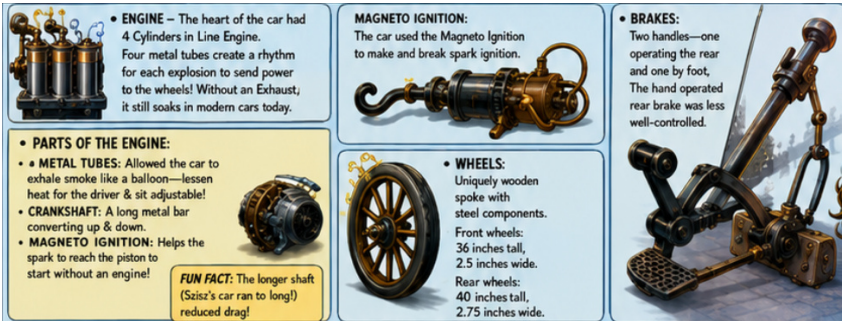
• **Honeycomb Radiator:** Bees living in the engine! No, it is the inner cooler of the car. When the engine heats up, there is water near the engine to cool it down. Then the water is sent to the honeycomb radiator to cool it so that it doesn't evaporate. (**Fun Fact : The HoneyComb Radiator was made by Wilhelm Maybach**.)

• **Magneto Ignition**: The car used the Magneto Ignitor to make and break spark ignition .

2. **Wheels:** Usually the wheels were made with wooden spokes with steel components. **Did you know that the front wheels were 36 inches tall and 3.5 inches wide, and rear wheels 40 inches tall and 4.7 inches wide!**

3. **Brakes:** There were two brakes, one operated by hand and the other by foot. But the hand-operated main haft was water-cooled.

1950



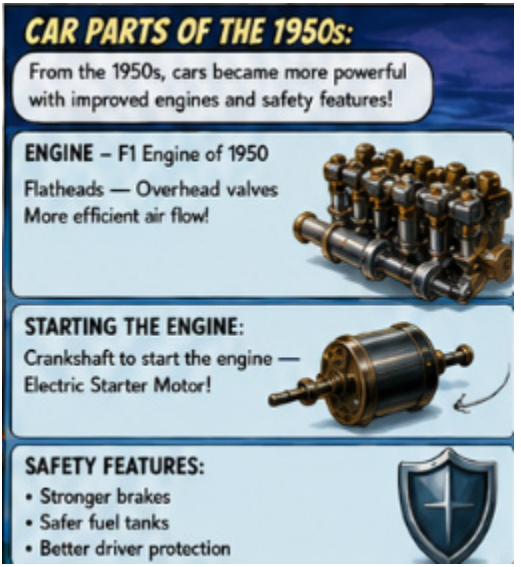
Past-O-Meter, "Transporting you to the championship of 1950." The date May 13th 1950 was the year in which the first ever international championship race was conducted, and the F1 World Championship was created with a boom. Did you know the race was held at the Silverstone circuit (Fun Fact: The distance is 4.649 km and there were 70 laps) in Northamptonshire, UK! Until now it was just car racing history; now it is F1 history.

Did you know that the first champion was Giuseppe "Nino" Farina, who, when aged nine, started to ride a two-cylinder "Temperino"! Amulya particularly likes this era, because this is the start of F1 and she loves the F1 sport. The championship included 6 major European Grand Prix and one American Prix (Britain, Monaco, Switzerland, Belgium, France, Italy, Indianapolis 500)

1958

Past-O-Meter ,"Transporting you to 1958" This year, the championship title was won by Mike Hawthorn. This was also the year F1 lost phenomenal racers Luigi Musso, Peter Collins and Stuart Lewis-Evans, Pat -O - Connor .This is a sad bit of F1 ! The year was tragic but the season was an interesting one. The year was the bridge of the old and modern era. Did **you know that the first female racer Maria Teresa de Fillipis started her Formula 1 career in 1958!**

Car Parts of the 1950s



Car Parts of the 1950s: Since we understood the 1900 cars let us see the powerful changes in the car from the engine to safety features.

Engine : From the 1900 the car became more powerful and the engine designers were superconfident of their design . Let us see the parts of the F1 Engine of 1950 that has evovled :

Flatheads - Overhead valves : The flatheads used in the 1900 were very inefficient for it to function. The Overhead valves were more efficient for air to flow directly through the cylinders.

Starting the Engine : In the 1900s there was Crankshaft to start the car which was exhausting and dangerous for the driver whereas in the 1950 the electric starter which was less dangerous and exhausting.

Cooling the water: (Ice Cooling time !) In the 1900s as you read in the previous pages they used the honeycomb radiator which was made of simple copper tubes. Whereas the 1950 used Dry Sump Oil Systems helped in increasing horsepower and better lubrication and Closed Cooling System which lubricated and cooled the water .

Supercharger: In the 1900 cars sucked any air around there whereas the 1950 used mechanical pumps which double, triple the power generated .

Suspension : Independent form of suspension was used. Transverse leaf springs **(not the actual leaf but its one long strip of metal running through the axle accessing suspension)** or coil springs were used.

Steering: Rack and Pinion System This system converts one kind of motion to another using a circular object(Pinion) and long , toothed bar (Rack) . **(I kind of like the way it sounds rack and pinion)**

Brakes: Drum Brakes **(Fun Fact: It is known as the drum brakes because of its hollow and cylindrical shape)** were mostly used but it was a refined version.

tires: Extremely narrow tires with High Aspect ratios (it means it is way taller than it is wide) placed on wire spoked wheels .

1960 - Golden Era Of Formula 1!

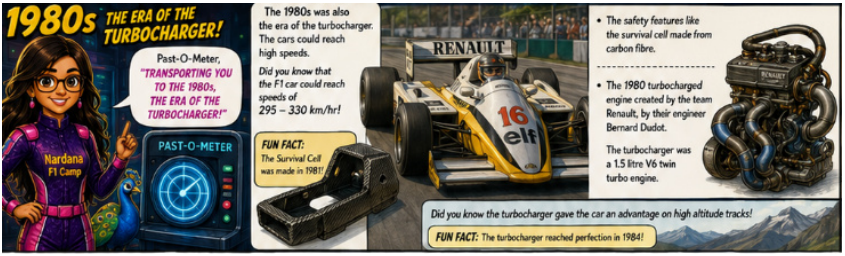
Past- O - Meter , "Transporting you to the 1960s, the golden era!" The era lasting from 1960s - 1970s was very glamorous yet very tragic . The Lotus team won their first Formula 1 trophy by Stirling Moss in Monaco in 1960. With the emergence of Aerodynamics by the Lotus cars with ground effect optimization , creation of the vacuum all of this started out with the Lotus 78 and later advanced to 79. The era was tragic. We lost 14 drivers in the 1960s and 12 in the 1970s. Did you know, in this era the safety of the F1 driver advanced from not wearing seatbelts to fireproof suits , helmets and harnesses! (Fun Fact: the first turbocharged car was created by Renault in 1977)

1980s - The era of the turbocharger

Past- O - Meter, "Transporting you to the 1980s, the era of the turbocharger." The 1980s was also the era of the turbocharger . The cars could reach high speeds.

Did you know that the F1 car could reach speeds of 295 - 330km/hr ! Safety features like the survival cell made from carbon fibre were made in this era.

(Fun Fact: The Survival Cell was made in 1981) The 1980 turbocharged engine created by the team Renault , by their engineer Bernard Dudot. The turbocharger was a 1.5 litre V6 twin turbo engine. Did you know the turbocharger gave the car an advantage on high altitude tracks! (Fun Fact : The turbocharger was made advanced in 1984).



1990s - The era of the technological embark

Past - O - Meter," Transporting you to the 1990s, the era of the technological embark" The 1990s seasons included the tragic deaths of Ayrton Senna and Roland Ratzenberger, the rise of the great driver Micheal Schumacher. The Driver Aids were banned very soon because of the risk of accidents, reduction of driver's skills during races and their high costs caused the less funded teams difficulties in managing their funds . **Did you know that the rule was banned in the year 1994 when the first invention of the Driver Aids started in the 1990s!** The 1990s also advanced the safety of the sport. In 1994 the invention of a wooden skid block to provide more grip for the car prevented advantages. Also there were pit lane speed limits introduced for safety.

Lastly the circuits were redesigned to prevent any accidents.

3 major inventors in F1 industry

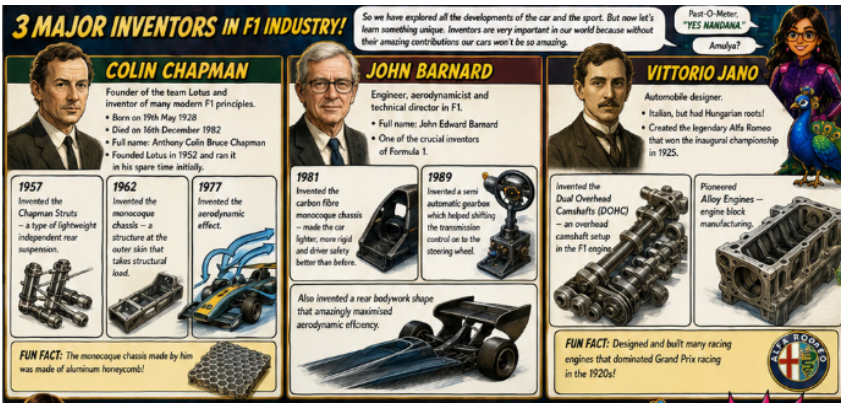
So we have explored all the developments of the car and the sport. But now let's learn something unique. Inventors are very important in our world because without their amazing contributions our cars won't be so amazing. Ready Past - O - Meter! Past-O-Meter,"Yes Nandana." Amulya? Amulya,"My favourite part , always ready!" Then let us explore major inventors of F1 !

1. Colin Chapman: The founder of the team Lotus and inventor of many modern F1 principles. He was born on 19th May 1928 and died on 16th December 1982. Did you know that his full name was Anthony Colin Bruce Chapman! He founded Lotus in 1952 and ran it in his spare time initially . Under his leadership pioneered the Chapman struts (It is a type of lightweight independent rear suspension.) as rear suspension devices in 1957 . A little later he invented the monocoque chassis in 1962 (a structure at the outer skin that takes structural load). Fun fact: The monocoque chassis made by him was made of aluminum honeycomb.

He even invented the aerodynamic effect in 1977.

2. John Barnard : He was an engineer , aerodynamicist and a technical director in the field of Formula 1. Did you know that his full name was John Edward Barnard! He was one of the crucial inventors of Formula 1 . He was the father of the Carbon Fibre monocoque chassis. He invented this in 1981 . The material helped in making the car lighter , more rigid and made driver safety better than before. He made a semi automatic gearbox invented in 1989 , which helped shifting the transmission control on to the steering wheel. He even invented a rear bodywork shape that amazingly maximised aerodynamic efficiency .

3. Vittorio Jano : He is an automobile designer. Did you know that he was Italian but had Hungarian roots! He created the legendary Alfa Romeo that won the inaugural championship in 1925 . He invented the Dual Overhead Camshafts (DOHC) , an overhead camera setup in the Formula One car. He also pioneered Alloy Engines engine block manufacturing.



Jokes

1. Why did the 1980 car time travel to 1990?

• Because it wanted an upgrade!

2. What did the supercharger tell the 1901 engine?

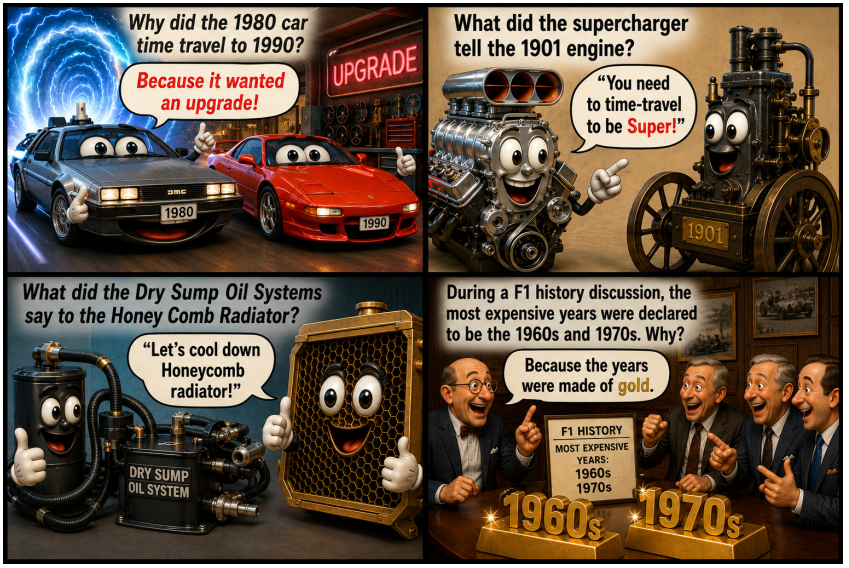
• " You need to time -travel to be Super!"

3. What did the Dry Sump Oil Systems say to the Honey Comb Radiator ?

• "Let's chill Honeycomb radiator!"

4. During a F1 history discussion , the most expensive years were declared to be the 1960s and 1970s. Why?

• Because the years were made of gold.



Quiz

1. What was the advantage of the turbocharger?

2. When did the Lotus team win their first trophy?

3. When did F1 racing begin?

4. When was the FIA created and what is the full form

5. When did Driver Aids get banned?

6. When was the turbo charger created?

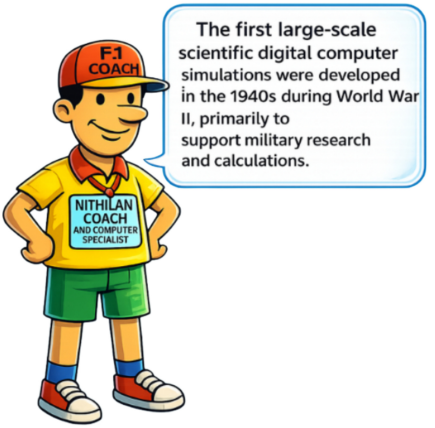
CHAPTER 03

CODE CREW

Hi, I am Nithilan, the Formula Racing Camp Coach and also a Computer Specialist. Welcome back to the present! Now that you have gone far in the past of F1, let's go to the futuristic parts! The future and present of technology is understood by the past. In Formula Racing Camp, we use computers a lot, especially Pro-Comp-So-Rotatus. He loves giving little titbits of information.

Tech Specs - RAM

The F1 car's computer has a small but powerful specialized memory of RAM (Random Access Memory).The ECU and car sensors monitor engine, throttle, temperature, battery, and many other systems; teams also analyse weather and wind data separately. That Ram sure gobbles a lot of memories.



Simulations

The main use of computers in F1 racing is doing simulations. A simulation is an online model of something (in this case, an F1 car), to test how it works, or to predict events. Looking at an F1 car, you might think the shape is weird, so long and wide, but less than a meter tall. But the shape of the car was carefully designed by computer simulations to be very fast. F1 cars are very expensive, so instead of crashing real cars for every small modification to test durability, the manufacturers simulate a crash to test how resistant the car is. The drivers can crash virtually and keep their jobs. The simulations can also find out the best time to pit stop, and the best materials to use in the car and tires. This is done by doing the race on the computer thousands of times, with different pit-stop timings and car materials to see which one works best. In the 2017 movie Cars 3, Lightning McQueen (Owen Wilson) and the Jackson Storm (Armie Hammer) use a race simulator to practice racing without going on the track and also comparing with competitors' speed and method. Similarly, drivers use simulators to learn racing in new tracks, saving time and money and also reducing crash-risk during practice. In fact, the FIA runs esports, or virtual championships.

Live simulations are also run, with factors such as windspeed, weather and competitors' speed to determine the live best way to race.

Steering Wheel

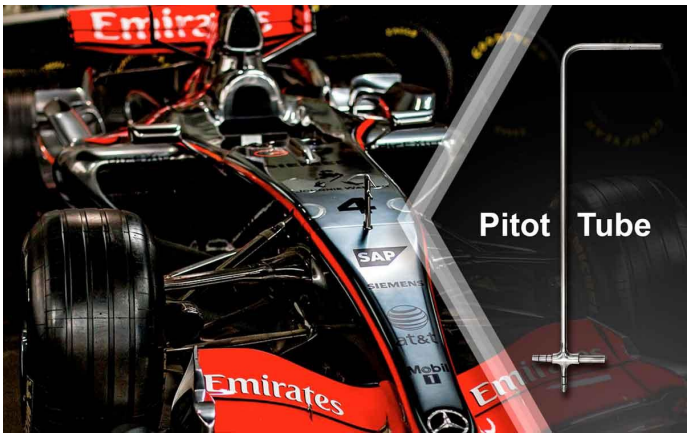
Surprisingly, the steering in an F1 car is basically a powerful computer. Like in some modern taxis, there are screens on the steering wheel to display data (In team Formula Racing Camp, we have Pro-Comp-So-Vehiculus in the cars). The computers process data like temperature, fuel, battery use, etc, and display them on the screen. (We in Formula Racing Camp also send the data to Pro-Comp-So-Rotatus for simulating). The steering wheel has 20-30 buttons, switches, knobs and dials to control engine speed, talk to the coaches, etc.

Telemetry and 5G

Did you know, F1 helped push the need and increase popularity for faster data systems like 5G?

We learnt that F1 Cars are basically high-speed computers on wheels. A lot of data is sent from the car to off-track computers in a process called Telemetry. One of these sensors is in the image here.

Air sensor: The Pitot Tube



The Pitot Tube is the sensor that tracks air speed and pressure. Engineers use this for designing with aerodynamics. It has a hole in front from which air enters.

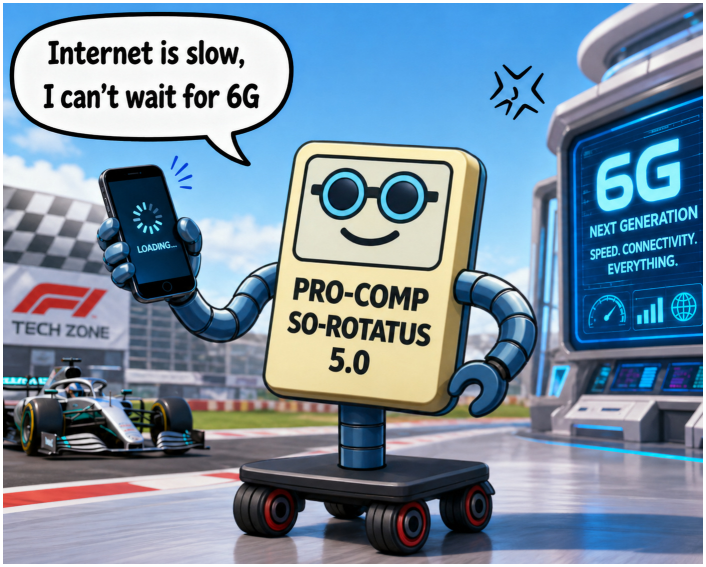
Data Points

We use **Data Points Per Second** (DPPS or just PPS) to describe the data. This is similar, but not to be confused with GB or some other bytes per second. GB per second is the amount or storage of data moving from 1 computer to another. Data Points Per Second is the amount of times the data is refreshed and sent.

For example, if a color sensor sends data at 2 DPPS, then one feed of data comes 2 times in one second. F1 Computers send data at **millions of DPPS!** Engineers need the data updated so fast because the cars go so fast. They move at more than 300 km/h. So they move ~80 metres in one second! Even if they delay ⅒ of a second, they can move 8 metres, potentially crashing or missing the pit box. So engineers need to constantly be aware of information such as tire wear, engine temperature, motor condition, etc. Now you must be wondering how phone networks are related to all of this! Well, more than a million data points per second is not easy. In the mid-2010s, **more than a million DPPS** was needed and only 4G and radio was available, which could work, but were stretched to their limit. So in 2017-19, F1 partnered with AWS (Amazon Web Services) because fans wanted stats and in 2019, they partnered with telecom companies. They started testing out 5G on F1 cars for **faster Telemetry**.

What is 5G and 4G anyway?

5G and 4G stand for **5th Generation and 4th Generation**. They are basically different versions of internet connection with different download rates and internet speed. 4G download rate is ~2.5 MB/s and there's a 30-50 millisecond delay in loading a page in optimal conditions. 5G download rate is ~25 MB/s and there's usually only around a 10 millisecond internet delay.



Fun fact: On 4G, it takes roughly 7 minutes to download a normal high definition movie, whereas on 5G, it takes only 6 seconds!

When F1 started testing and exploring technologies like 5G for faster data communication, telecom companies' 5G also gained more attention and more customers bought 5G phones.

Computers in Pit stops

Computers are also extensively used in pit stops. Pit stops are when the car enters the pit area for changing wheels or other issues. They are, of course, used to calculate and simulate the car stopping exactly in place. But they are also used in the pit machinery -

Wheel guns

The pit crew uses high speed drills to secure the bolts on the wheels. There is a small sensor connected to a computer on the wheel gun which can sense the tightness of the bolt. The sensor sends a signal and prevents the car from moving until all bolts are proper. All of this happens within a fraction of a second, showing the speed and precision needed in professional F1 Racing.

Cameras and sensors

There are numerous cameras in and around the pit which monitor incoming cars and also sensors which monitor the condition of the car in the pit. They are used to prevent crashes by showing signals like a traffic light. Red for wait, green for go. This replaced the earlier system with a lollipop signal.

Collecting data

Apart from on-track Telemetry, during the pit stop, cameras and sensors collect data about the car's functions. This data is sent to the team the driver is racing for. This can help when testing new tires for example.

Communication

Computers are used for communication between the pit crew and driver. For example, "Gearbox is rusty, change gears slowly"

All of these are crucial for correctly executing a pit stop.

Professional pit stops usually take less than 3 seconds, with the world record at 1.80 seconds, achieved by Lando Norris at the 2023 Qatar Grand Prix for McLaren.

Safety Systems

The most essential, life-saving use of computers in F1 is in Safety Systems.

Computers are used in these safety systems -

Accident Data Recorder (ADR)

This one does not save lives directly, but prevents crashes in the future. If a crash happens during a race, then moments before and after the impact, data such as speed, braking, throttle, and steering is sent on the Telemetry and is later analyzed by FIA.

Car-Health Monitoring

The data sent on Telemetry during a race is analyzed then and there. The race engineer gives instructions on how to drive better. The performance engineer analyzes data to store lap speed and turn speeds. Systems engineers are the most important for safety, they analyze the health of the car and decide if something is wrong.

Automatic system shutdown

If the danger is too high, then subsystems such as a particular voltage unit shutdown and warning lights flash.

Track wide safety systems

Cameras and sensors around the track track car speed, steering, etc and if abnormalities are detected, warning lights and flags flash on steering wheels and around the track as seen in this image. Yellow means that danger is ahead, such as debris from a crash ; the drivers must slow, but can continue racing. Green means the danger is over ; it usually succeeds yellow lights. Red means that the entire race is stopped, usually because of a major accident ; the drivers must return to the pits. Blue comes for a slow driver during the race ; it warns them that faster drivers are behind and must allow them to pass. White means that a service vehicle, such as a Mercedes Benz safety car, is ahead ; the driver must be cautious.

Biometric monitoring

F1 biometric gloves were introduced to help monitor pulse and blood oxygen/pulse oximetry, not blood pressure. . If the drivers have a health issue, they slow and come to the pits. The FIA has mandated usage of this since 2018.

Broadcasting

Not everyone goes to a stadium to watch F1, many people prefer watching it online from many views and angles. The FOM (Formula One Management) is the center of broadcasting F1 racing. They produce a single global video feed called the **World Feed** which has shots from all angles. The broadcasting companies bought this and added their own audio (commentary and music). Before the 21st century, television companies had their own cameras. This was discontinued because of bias and poor quality because companies from one country are likely to be partial to their country and also keep showing shots of a team popular in their country.

World Feed

• Shot by FOM.

• Bought by streaming companies as part of an F1 rights bundle which can cost from $5 million in India to $250 million in Great Britain.

• Streaming companies add advertisements & commentary.

F1 TV

• The Formula 1 group also releases their own live stream of the F1 races.

• They release the world feed to consumers on their own OTT.

• Consumers can choose between many views, beside the main view. This includes track views and also views inside the car.

Jokes

Q. Why didn't the F1 computer crash?

A. It had already passed the crash test!

Q. Why did the F1 computer bleat?

A. It had too much Ram!

Q: Why does an F1 steering wheel have so many buttons?

A: So the driver can press the wrong one in style.

Q. Why do F1 drivers use simulators?

A. So that they can virtually crash and keep their jobs.

Q. Why was the wheel gun so confident?

A. Because it knew how to handle pressure.

Q. Why did people never argue with F1 Data?

A. Because the data always had a point.

Quiz

Q. What were the first computer simulations for?

_____
_____.

Q. How much RAM does an F1 car's computer contain?

_____.

Q. What is the term for sending data wirelessly from the car? _____.

Q. Which sensor in an F1 Car tracks air pressure and speed? _____

Q. What is the download rate for 5G? _____MB/s.

Q. 5G stands for 5th _____.

Q. The _____ engineer analyzes lap speed and turn speed.

Q. A professional pit stop takes less than _____ seconds.

Q. DPPS stands for _____ Points Per Second

Q. Apart from real crashes, teams use _____.

CHAPTER 04

QUANTUM 1

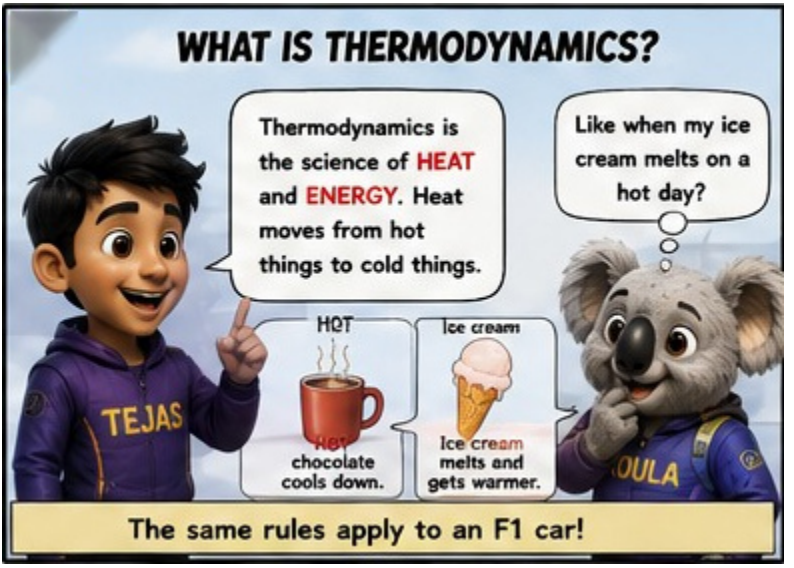
Hey campers!

I am Tejas, an F1 racer, and today I will take you for a trip around the world!

Formula 1 is the most loved sport due to its phenomenal speed, overtaking maneuvers, and nail-biting-edge-of-the-seat finishes. This is the reason why I am fascinated and dragged towards Formula 1 too. Another interesting aspect to this sport is the **Physics,** which is quite intriguing.

All of you join me in the journey into the world of **Thermodynamics**. It may sound complicated, however, in reality it is very simple. It is the science of **heat and energy** (the backbone) of Formula 1.

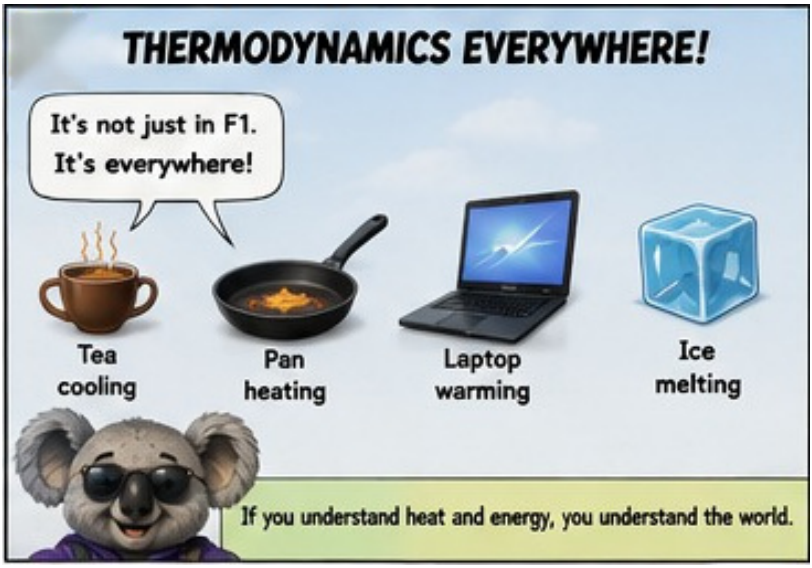
Thermodynamics



Imagine a Formula 1 car having a straight line speed of over 300 km/h which only indicates how the engine needs to perform to handle such breakneck speeds. The tires and brakes of the car are ever/ready for the turn at chicanes. All these components produce heat.

Heat Control (overheating) is the vital aspect that needs to be regulated by the teams. The engineers monitoring the car help in regulating the temperatures to keep running the car and provide necessary instructions to the drivers on overheating. At times, overheating causes braking issues. In a similar vein, cold temperature has adverse effects on the performance of the car.

Think about this: when you run as fast as you can for one minute your body gets warmer because it is using energy. Similarly, Formula 1 cars use a lot of energy producing a lot of heat.



In this chapter, we will learn where the heat in a Formula 1 car is derived from, its movement and how important it is to regulate the heat factor during the race.

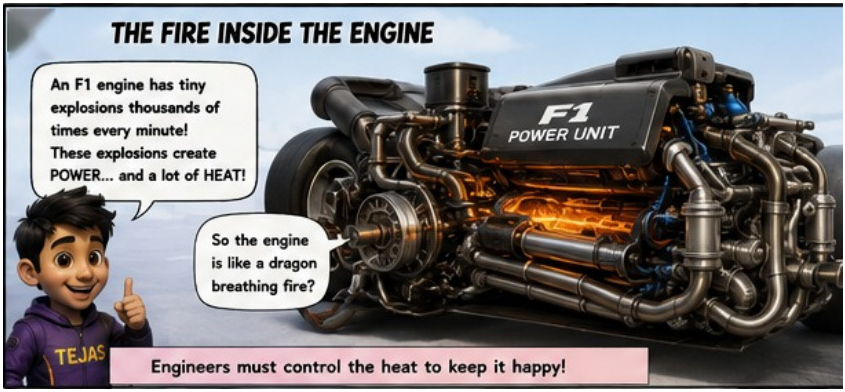
The Engine (Again…)

Let us start with the heart of the car. The engine produces over 1000 horsepower powered by 1.6-liter turbocharged V6 hybrid engines.

The engine is a powerful thing.

It is like an explosion that happens over and over again.

The engine is an extraordinarily intricate piece of machinery in FFormula One racing, with tiny explosions.



The engine mixes fuel and air together. Then these tiny explosions happen which create a lot of energy, some of this energy takes the car forward and the thunderous sound that we can hear.

A lot of the energy generated from the engine can be fatal when overheated which should be taken care with utmost caution.

If the engine gets too hot, the parts inside the engine can get worn out and eventually break.

The teams that work with these engines use radiators and airflow systems to remove the excess heat from the engine.

Imagine you are wearing a thick winter jacket on a really hot summer day, not only would you feel really uncomfortable, also unable to keep yourself cool.

The engine has a problem with overheating.

That is why cooling the engine is just as important as making the engine powerful.

The power unit provides the energy that helps the car accelerate and perform on track.

So they make sure to take care of the engine.

Imagine you are driving fast at 320 kilometers per hour and you see a corner coming up.

What do you do in this situation?

You use the brakes to slow down.

The brakes have to slow the car down quickly.

When you press the brake pads against the brake discs it makes friction.

This friction makes a lot of heat.

Sometimes the brakes get so hot that they are bright orange in the dark.

The temperature can get really high, over 1000 degrees Celsius!

If we did not have friction, the car would be impossible to stop.

If we did not have brakes, it would be impossible to have racing.

Next time you ride a bike try touching the brakes after you have been riding.

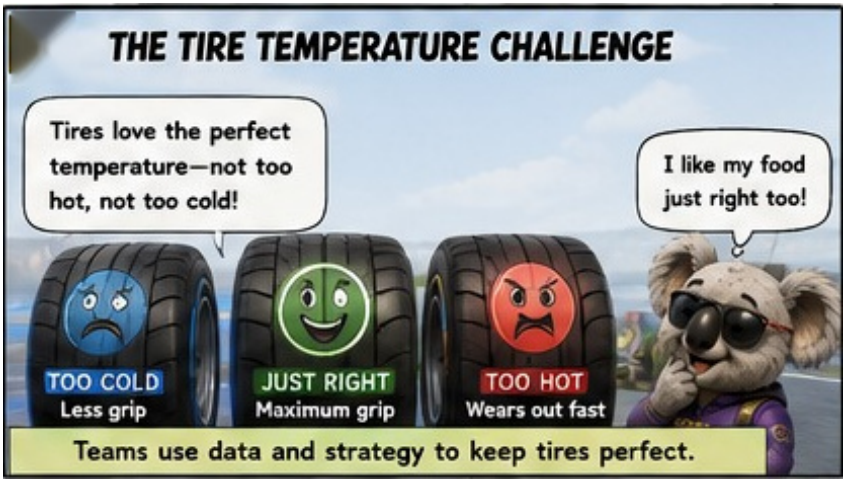
You might notice that they feel warm.

This is because the same thing is happening with the brakes on your bike as it is with the brakes in Formula 1 it is just happening faster.

Tires(Again…)

Tires are super important for an F1 car.

The tires are the parts that touch the track surface directly.



If the tires are too cold they do not grip the track well.

If the tires get too hot they wear out fast.

Teams try to keep the tires at the right temperature.

Drivers move around a bit before races to help warm up their tires. They also use blankets to get the tires close to the perfect temperature before they start driving.

Imagine running on ice wearing shoes that're too slippery. That's like driving with cold tires.

Now imagine running with shoes that are melting and falling apart. That's similar to driving with tires that're too hot.

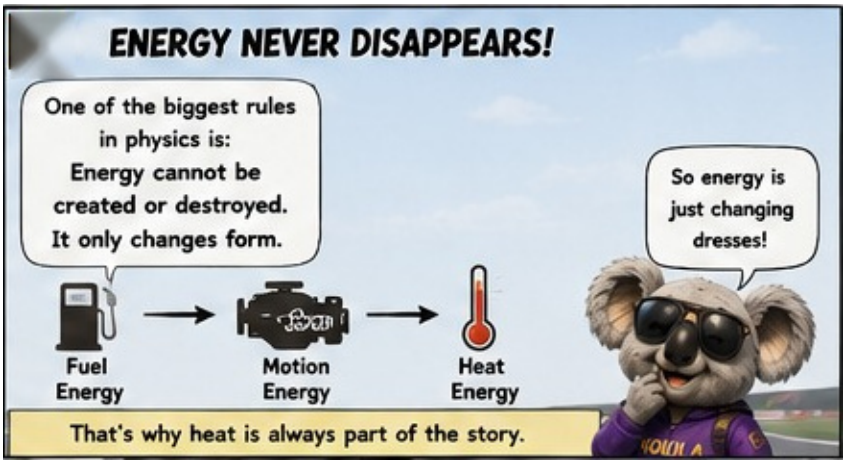
The best performance comes from having the tires at the right temperature.

Drivers often complain about tire temperatures on the radio because they want to be sure their tires are in the best possible condition. They know that if the tires are too cold or too hot, it will affect how well they can drive. Getting the tires into the right temperature window is crucial for performance. There are times when it rains on the track, the tire needs to be changed accordingly, else the car will not be able to perform and slip and slide.

Energy!

One of the important ideas in physics is that **energy cannot just vanish**.

Energy cannot be destroyed.
Energy changes form.



For example, the energy from fuel (chemical energy) which becomes movement (kinetic/mechanical energy). Then this movement becomes heat (thermal energy). This heat then moves into the air (dissipation). All of this is interconnected.

When an F1 driver speeds up, the energy from fuel is changed into movement. Some of this energy from fuel also becomes heat inside the engine. the tires and the brakes.

Physics is really the story of energy, fuel changing from one form to another.

Aerodynamics (Again…)

When a Formula 1 car is moving, it has to push the air out of its way.

As the Formula 1 car needs to move fast, it has to fight really hard against the air. This is called drag.

The drag on a Formula 1 car slows it down. Makes the engine work a lot harder.

When the engine of a Formula 1 car is running it gets really hot.

The engineers and the team who design Formula 1 cars spend a lot of time making sure they can cut through the air easily.

Every single part of a Formula 1 car like the wings and the curves and the surfaces is there, for a reason.

Air is around us. We can not see it but it has a big impact on Formula 1 cars.

Formula 1 cars are really affected by air.

Drivers are athletes; they need to be in shape to drive these cars.

It gets really hot inside the cockpit.

The temperatures can get extremely high when they are driving.

To protect themselves drivers wear suits, helmets, gloves and other safety equipment.

When they are racing they have to steer the car, brake, accelerate and stay focused, for a time.

All this hard work makes their bodies hot. This heat makes them sweat. They lose water. So drivers need to drink water to stay hydrated.

If a driver is tired they can make mistakes. If a driver is cool and focused they can drive really well. The way heat and energy work, which is called thermodynamics, affects the car and the driver.

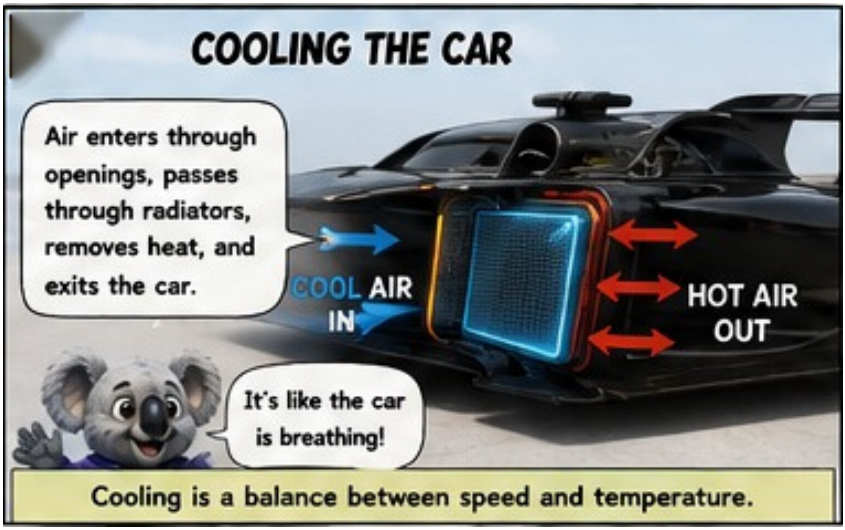
Thermodynamics affects not the car but also the person driving the Formula 1 car.

Drivers can lose 3 kilograms during a hot race weekend, which is a lot of weight to lose in just a few days.

Back to Heat

If the car gets too hot, there will be problems.

The car needs a cooling system.



Air goes into the car. Then it goes through the radiators. The radiators get rid of the heat that the car does not need. The hot air is then taken away from the car.

This happens over and over during a race.

Cooling the car is a vital important factor.

If the car has big openings to let air in, it can make the car slower because of the air pushing against it.

The people who build the car have to figure out what is just right, for the cooling system of the car.

Physics

Congratulations, Camper.

You finished Quantum 1: The Physics of Speed.

This is a big deal.

You learned about thermodynamics which is the science of heat and energy.

You found out how engines make heat and why brakes get so hot, they glow orange.

You also learned why tires need to be at the temperature and how drivers stay cool when it is hot.

You learned how engineers control temperatures during a race.

The important thing you learned is that Formula 1 is not just about racing.

It is like a laboratory that is moving really fast, over 300 km/h.

Every time the cars go around the track we learn something about energy and heat and motion and science.

As you may have noticed, Physics is interconnected with everything else, biology, chemistry, design and mechanics and of course, Mathematics.

When you watch a Formula 1 race do not just watch the cars go fast.

Look for the physics that's behind it, the physics of Formula 1.

Now, have some fun:

Jokes

1) Why did the F1 tire go to school?
Because it wanted to improve its **traction** in class!

2) Why don't Formula 1 brakes ever tell secrets?
Because they always **spill the heat**!

3) What did the engine say after winning the race?
"I'm exhausted, but I'm still fired up!"

4) Why was the Formula 1 car so good at science?
Because it had an excellent **drive**!

5) What happens when an F1 car tells a joke?
Everyone laughs at **top speed**!

Quiz

1) What is **Thermodynamics**?

A) The study of plants B) The study of heat and energy

C) The study of oceans D) The study of stars

2) What creates heat when two surfaces rub together?

A) Gravity B) Friction C) Magic D) Magnetism

3) Why do Formula 1 brakes become very hot?

A) They are painted orange B) They convert motion into heat

C) They are near the engine D) They absorb sunlight

4) Which part of an F1 car needs the correct temperature for maximum grip?

A) Rear Wing B) Steering Wheel C) tires D) Helmet

5) What happens to heat production when an F1 car goes faster?

A) It decreases B) It disappears C) It stays exactly the same

CHAPTER 05

ECO RACING

Hello campers!! I'm Nihira – a true F1 driver and nature lover. I'm accompanied by Chicane. She's a high speed eco-warrior penguin. I first met her during my voyage to Antarctica, back when she was an out- of -place penguin who dreamt big and hated the cold (why is she a penguin then ??). Luckily, I rescued her from her 'absolutely fine' home, and now she's living the time of her life.

I also have an ultra techy and highly exclusive device.

Introducing VORTEX, which can take you anywhere. Campers, this device is crucial to help us teleport – make sure you remember it.

It has some funny rules that will be unraveled throughout the journey

"Where are we heading?", you might ask?.

We are going on a behind-the-scenes tour. That MAY OR MAY NOT sound boring, but trust me, you are about to witness the wonders of how F1 saves not only our lack of thrill in life, but also the ACTUAL life on Earth. They aim to achieve Carbon net zero emission by 2030.

Green Machine

We are now in the Centralised Power Generation Compound, placed right outside the perimeter of the paddock.

A looooooong word for a generator.

But not just any generator, it keeps the ENTIRE F1 event running, so we are obliged to love it because otherwise, you won't have F1, and that is BAD.

These generators are made by a company called Aggreko. They're compact, don't make much noise and the best—they're eco friendly.

They run on a fuel called HVO (Hydrotreated Vegetable Oil) , which is also known as renewable diesel.

They sit right outside the paddock, silently doing their big job.

These good times were bestowed upon us not very long ago. The 2023 Austrian Grand Prix was the first time this generator was used, but it was widely accepted only by 2025!

Earlier, there were big, old smoky diesel trucks. Back in the day, the generators were placed on HUGE 18 wheeler trucks.

They brought along dust smoke and harmful fumes with them, which flooded the paddock. That is freaky, because the VIPs, and most importantly, drivers like us, sit in there, and it is definitely not fun to catch a cold a day before the race (I talk from experience)

Ok ,I think that's a looooot of information. I guess we deserve a joke break, also known as a Jerak, which is what Chicane and I call it.

Now these jokes are provided by VORTEX. I hope you campers remember that this device had some weird rules. Well, here's rule number 1 – VORTEX gives you 2 and only 2 jokes after every bit of learning.

That way, you can stay ON TRACK (get it?)

JERAK TIME

Why did the mechanic get a punishment from the team kitchen?

He put olive oil in the generator and made the pit lane smell like garlic bread!

Why did the team principal bring salad dressing to the generator room?

He heard that the new fuel was made of plants and wanted to fix the smell!

Wow! I always love the jokes VORTEX cracks. It always gives jokes which are a hybrid of Formula 1 and classic humour.

Hey what's that word I used– hybrid? Well, hybrid in simple terms means a mixture of 2 things. Something made of 2 elements. You might have heard of hybrid cars before. The concept is that the power of the car is divided between petrol and electricity.

Even F1 cars' engines are hybrid. Let's go and see what actually happens inside.

Hybrid engines (again)

Formula 1 engines are now officially called power units. The engine power split is nearly 50-50 now, between electric power and the combustion engine.That wasn't there until 10 years ago. Till 2015, the power was divided with only 20%

distribution to electric power. Since the Formula 1's goal to reach Net Zero

Carbon by 2030, they have increased it to 50-50. Now you may wonder, why did they not make the cars completely electric? Why stop at hybrids?

There is actually a very good reason why they can't go electric. The current battery technology lacks the energy density to support extreme speeds and a power output to support a full 305 kilometers of the grand prix.

Even though Formula 1 can't go EV, that doesn't stop them from making sustainable fuels that reduce carbon footprint even more. The fuels are designed from carbon capture, municipal waste and non food biomass.

Again, these good times weren't always around. In the 1950s, the fuels used were HIGHLY TOXIC chemical cocktails. Methanol and Ethanol were used as coolants. Benzone andBenzene was added for power, but they were highly carcinogenic.

Then we had Nitrobenzene, a highly volatile and potentially explosive compound, which was added to the mix. Chicane almost fainted hearing that.

In 1958, the FIA finally decided to ban this alcohol mixture and instead replaced it with

Aviation Gasoline (AVGAS).

There were immediate impacts because of the ban:

Huge Power Drop: Teams like Vanwall and Ferrari instantly lost about 20 to 30 horsepower because they could no longer use their high-energy alcohol mixes.

Melting Pistons: Because alcohol burns much cooler than gasoline, engines suddenly run much hotter. Teams spent the entire 1958 season redesigning their engine cylinders and pistons so they wouldn't literally melt from the heat of the AvGas.

A Level Playing Field: It stopped teams with better chemical labs from having an unfair advantage, forcing everyone to compete using the exact same base fuel.

With so many issues, AvGas only lasted until 1960.

In a desperate effort to make it better, FIA decreed that starting in 1961, F1 cars would use normal commercial grade fuel. Literally the same petrol you would get at a roadside gas station.

On the track, 1960 marked the absolute death of front engined F1 cars. Cooper-Climax (A legendary British racing car that revolutionised F1 in the late 1950s), had dominated the 1959 and 1960 seasons by putting the engine behind the driver.

Seeing how much faster and nimbler these lighter cars were, virtually every team spent 1960 abandoning their front-engined designs. Slowly things changed, from toxic alcohol mixtures to commercial petrol and dangerous, uncontrolled turboengines in the 80s to hybrid turbo engines today. It certainly has been a long journey.

Well, let's see how much you've learnt. Rule number 2 for Vortex - It gives you 3 questions and see if you've paid attention (teacher attitude)

Quiz

1) What is the goal behind Formula 1 going green?

_____

__

_____

__

2) What is the name of the company from which Formula 1 buys the generators? _____

3) Formula 1 has transitioned to using sustainable low-carbon generators in the paddock. What primary energy source and fuel type are used to power these advanced units?

_____

_____

_____

CHAPTER 06

MAP MASTERS

Formula racing is always known for being the fastest circuit sport in the world. What few people know about F1 racing is that it is among the geographically diverse sports ever created. In every competition, the location is completely new and unique, and it changes everything from how the car runs to how the drivers drive and even how the races are conducted.

Every race is not only against the competition; it's also against nature. Everything, from the ground the car drives on to the air, the weather, and even the streets of the city, all contribute to how the game turns out.

We have gone through history, into the future, and seen the now.

Today we go deeper into… LOCATION

And in Formula 1 racing… the world itself becomes your racetrack.

Seas & Mountains: Monaco

Welcome to Monte Carlo, the place where the famous Monaco Grand Prix takes place.

Monaco can be called a unique destination in Formula 1 racing since this destination is located in between steep hills and the beautiful Mediterranean sea coast.

The Monaco track is not a track created specifically for races; it is made of the streets of the town itself, which means that there cannot be any mistakes while competing here.

Driving around this circuit requires a lot of skill due to small roads, difficult turns, and lots of changes in elevation, but we shouldn't forget about challenges such as bumpy driving surfaces, building shadows, and fast changing illumination (when leaving a dark tunnel).

The Fairmont Hairpin is one of the slowest turns in Formula 1, and still The precision in maneuvering is essential while going through it.

This location is special not only due to the difficulty of the circuit but also the surrounding atmosphere as well.

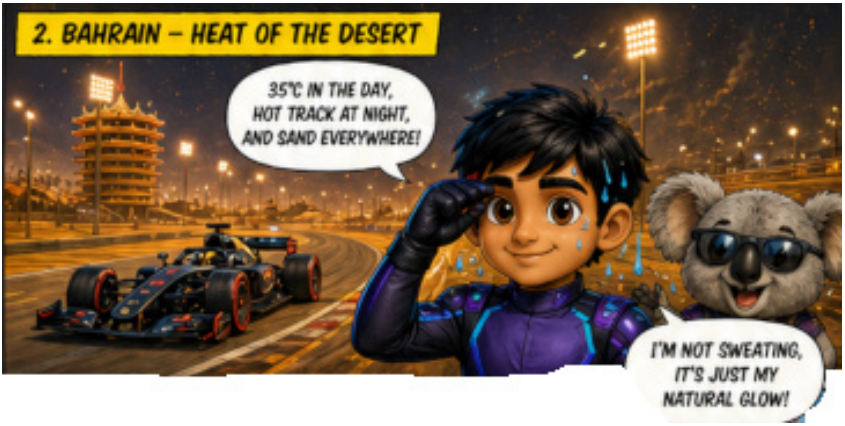
The Bahrain International Circuit

The Bahrain International Circuit combines the elements of the desert into the race itself.

This circuit is based in the Middle East and finds itself surrounded by the dry and sandy land that is characterized by hot temperatures during the daytime. The racing event takes place at night time, but this does not make the surface cool since the accumulated heat stays on the asphalt.

One of the aspects that should be taken into account while driving through such conditions is that the asphalt becomes heated due to the sun, which makes tires prone to overheating and losing grip.

Moreover, another important element is that sand can blow into the racing track, affecting the braking distance and grip.



Dry air affects the engine and cooling system, and therefore careful calculations should be made by teams.

Circuit de Spa-Francorchamps

Welcome to Circuit de Spa-Francorchamps , one of the most legendary tracks on the planet!

The circuit is notorious for its long distance, varying elevations, but above all else, for the capricious nature of the weather conditions there.

Due to being surrounded by forests and hills, the track is subjected to its unique weather phenomenon.

It is quite possible that one part of the circuit will be bone-dry while others might be under heavy rainfall. This means that the driver is required to keep adjusting his or her approach during the race.

Perhaps the most well-known turn here is the Eau Rouge and Raidillon turn where the car ascends steeply up the hill.

Spa teaches every driver a valuable lesson:

Be prepared for anything!

Suzuka International Racing Course

Racing at the Suzuka International Racing Course 🇯🇵 is a true art of accuracy.

Suzuka is one of the rare circuits that form a figure-eight shape, as one part of the circuit passes over another using a bridge. Such a design gives rise to a rhythm that racers need to understand.

"Turns of S" require smooth driving and impeccable timing. Every step, whether braking, accelerating, or handling the wheel, must be calculated precisely.

Consistency and accuracy will get you through here. While street races offer speed, Suzuka gives it to those who remain balanced.

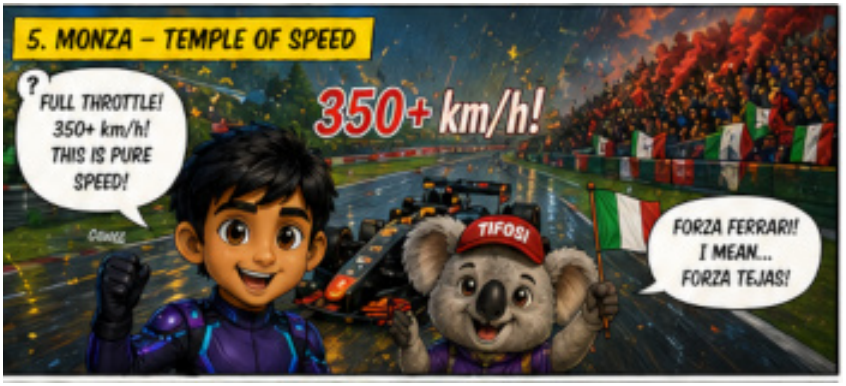
Passionate and respectful Japanese fans are what make the race even more exciting.

Suzuka is a track where:

accuracy is key.

Autodromo Nazionale Monza

Welcome to Autodromo Nazionale Monza 🇮🇹, also called the Temple of Speed.



Monza is one of the oldest race tracks in F1 and is well-known for its long straight sections and fast-paced racing.

Due to this track's long straight portions, slipstreaming becomes crucial for drivers. They try to drive close to another vehicle to minimize drag and increase their speed.

Nevertheless, due to its fast pace, the circuit has numerous braking sections. The drivers need to brake rapidly when approaching chicanes.

The energy from the Tifosi makes the atmosphere unique.

All efforts made by the drivers at Monza aim for one goal:

maximizing speed.

Marina Bay Street Circuit

The Marina Bay Street Circuit 🇸🇬 takes place during the night.

It was the first Formula 1 night race and continues to be extremely challenging.

The humidity and temperature make it physically demanding on drivers.

Drivers can lose up to 3 kg of their body weight as a result of sweating. It is a long race with a lot of corners that requires concentration.

The circuit itself is narrow and rough, making it difficult for racers.

The bright lighting makes reflections off buildings more visible.

Singapore pushes humans to their physical limits.

Autódromo Hermanos Rodríguez

Altitude matters at Autódromo Hermanos Rodríguez 🇲🇽.

With an elevation of more than 2200 meters, there's less oxygen in the air.

This results in lower engine power output and less aerodynamic downforce.

Circuits are slippery, braking is challenging, and there's less traction available.

Teams need to adapt their settings to account for low air density.

Fans in the stands in the stadium sector create an electrifying ambiance.

Here, geography influences even the intangible:

the air.

Albert Park Circuit

Race Dynamics at Albert Park Circuit 🇦🇺

Due to its proximity to the sea, the race track in question is affected by the dynamic weather conditions. Wind gusts influence car handling, particularly around fast corners.

Cooler temperatures make handling tires more difficult.

As the race weekend progresses and the number of cars on track increases, grip improves.

Drivers have to be aware of nature in Australia.

Nature cannot be seen—but it is always there.

Las Vegas Strip Circuit

At the Las Vegas Strip Circuit 🇺🇸, Formula 1 enters a new era.

Bright neon lights, long straights, and a city built for entertainment create a race like no other.the race here are less.

Buddh International Circuit

Welcome to the Buddh International Circuit 🇮🇳

Formula One in India is a thrilling combination of speed, tradition, and excitement. Situated in close proximity to the capital of India, New Delhi, the circuit is characterized by a wide design, high-speed sections, and smooth corners.

It is equipped with an extremely long back straight at which cars go at exceptionally high velocities, followed by aggressive braking zones. Drivers need to quickly switch from driving at full throttle to taking tight turns.

As far as the climate is concerned, India boasts a warm and dry climate that negatively impacts tire traction and causes accelerated tire wear. Similar to Bahrain, tire management is crucial.

What is more, there is one more feature that makes racing in India unique – the fans' enthusiasm. They scream enthusiastically and cheer every driver's accomplishment. You feel the power of the nation on your shoulders.

Experience:

The drive feels exhilarating and invigorating.

The track flows effortlessly, but requires utmost attention.

The heat challenges you, and the crowd inspires you.

This is where racing feels like home.

Koala:

Fast cars AND amazing food? I'm never leaving!

Travelling from one to another circuit shows how Formula 1 is geographically diverse. The world of Formula 1 is not only speed and the car but also the surroundings . Like how circuits in Monaco are different from the one in Suzuki .

The climates, the race tracks , the various hairpin bends to traditions and culture. This is me Tejas signing off…… Wait-Wait let's finish a Fun quiz and jokes at the end .

Quiz Section

1. Why is location important in Formula 1?
2. In which country is Monte Carlo located?
3. Which circuit is called the "Temple of Speed"?
4. What type of race is Singapore known for?
5. Which track has a figure-8 layout?
6. What makes Spa's weather unique?

Jokes

1. **Why don't F1 drivers get lost?**
   Because they always follow the racing line! 🏁
2. **Why is the Monaco Grand Prix so hard?**
   Because even Google Maps says: "Make NO mistakes!"

3. **Why did the car struggle in Bahrain International Circuit?**

Because it couldn't handle the sand-wich conditions!

4. **What did the driver say at Circuit de Spa-Francorchamps?**

"Do I need wet tires… or sunglasses?"

5. **Why do drivers love Autodromo Nazionale Monza?**

Because life is better in the fast lane!

6. **Why was the driver so tired in Marina Bay Street Circuit?**

Because even the air said, "I'm heavy today!"

CHAPTER 07

THE STRATEGY & PROFIT GARAGE

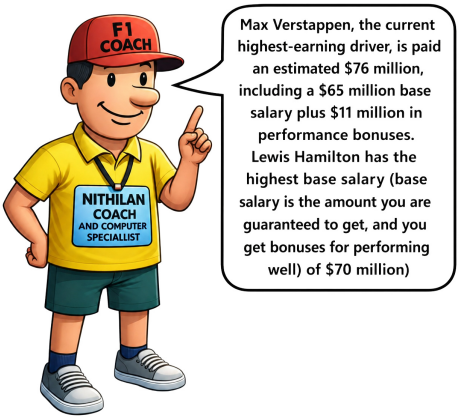
Welcome to the Strategy and Profit garage, campers! In this chapter we'll explore the administration of the FIA and FOM and the strategies used by famous F1 teams.

So let's start with a ride on the strategies F1 Car.

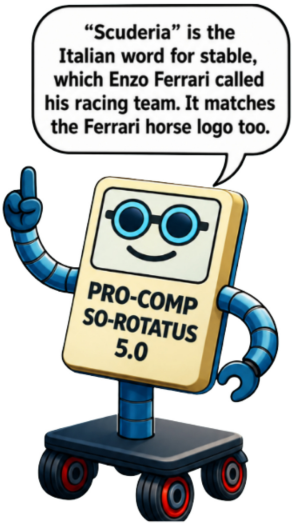
Strategies of F1 Teams

As of 2026, there are 11 F1 Teams. They are officially known as Constructors. Each team can only enter with 2 cars and drivers. The teams design their own cars following FIA regulations. The FIA also sets a spending maximum budget at around 215 million dollars.

This includes car design and building and salaries for engineers and mechanics but not salaries for driver and top managerial salaries, sponsorship, travel and accomodation.



Ferrari



First, let us look at the strategy of the most successful F1 Team.

Parts

Ferrari collaborates with Shell for fuel and fuel-related parts. In the 2026 season they are using ecofriendly bio-fuel made by Shell instead of crude oil.

They make their own engine and chassis from Maranello, Italy.

Sponsors

Ferrari's name sponsor is the computer company HP®, making their official name Scuderia Ferrari HP

Famous Drivers

Ferrari's most famous drivers are Alberto Ascari, Juan Manuel Fangio, and Michael Schumacher, whom you can learn more about in the last chapter.

History

Scuderia Ferrari was created on 16 November 1929 to enter amateur drivers into racing competitions by Enzo Ferrari, who was earlier a racer for Alfa Romeo. They participated in the first Formula One Championship season in 1950. Though they missed their first round in Britain, their best driver Alberto Ascari came 5th in overall standings.

Since then, they have participated in every season, and have won 16 World Constructor's Championship, 15 World Driver's Championship and a total of 240 Grand Prix races.

World Driver's Championship vs World Constructor's championship

At the end of each racing season, there are actually 2 prizes awarded, the World Constructor's Championship (above) and the World Driver's Championship (below). The WDC is for the driver with the most points and the WCC is for the team which has the highest total points of both (or the top-scoring in a race; the 2 drivers per team rule was only introduced in 1992) cars.

When we say someone won a Grand Prix, it means they only won one race out of 23.

McLaren Mastercard Formula 1

McLaren is statistically the second best performing F1 Team.

Parts

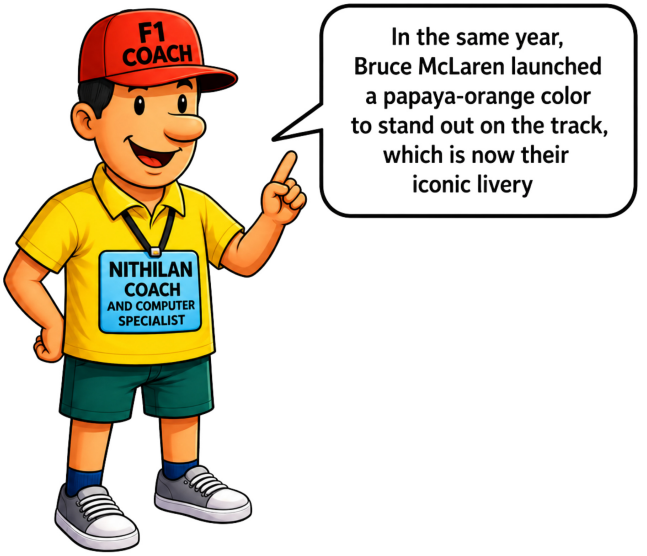
Their fuel is made by Petronas and transmission lubricants are made by Motul, as per their agreement for the 2026 season.

The McLaren team makes their own chassis and gearbox, but their engines are supplied by Mercedes-AMG.

The FIA mandates use of McLaren ECU (electronics control unit) for all the teams because they are robust and prevent teams from using banned electronics and promote fairness.

Sponsors

Their primary name sponsor is Mastercard, making their full official name McLaren Mastercard Formula One Team.



Famous Drivers

McLaren's most famous drivers are Lando Norris, Ayrton Senna and Lewis Hamilton.

History

McLaren was founded on 2 September 1963 by Newzealander Bruce McLaren.

Their first race in 1966 was a failure as the McLaren M2B car had to be retired because of oil leakages after only 9 laps

They won their first race at the Belgian Grand Prix in 1968, the racer being Bruce McLaren himself.McLaren won their first Constructors' Championship and Drivers' Championship in 1974, with drivers Emerson Fittipaldi and Denny Hulme. Fittipaldi won the Drivers' Championship, while the combined performances of both drivers helped McLaren secure the Constructors' Championship.

Teams Which Have Won Championships

RANK	TEAM	WDCs (Drivers' Titles)	WCCs (Constructors' Titles)	YEAR JOINED	YEAR LEFT
1	Ferrari	15	16	1950	–
2	McLaren	13	10	1966	–
3	Williams	7	9	1977	–
4	Mercedes-Benz	2, 7	0, 8	1954, 2010	1955
5	Red Bull Racing	8	6	2005	–
6	Brabham	4	2	1962	1992
7	Renault	2	2	1977	2020
8	Benetton	2	1	1986	2001
9	Brawn	1	1	2009	2009
10	Lotus	6	7	1958	1994
11	Cooper	2	2	1958	1969
12	Alfa Romeo	2	0	1950	1951
13	Tyrrell	3	1	1970	1998
14	Matra	1	1	1968	1972
15	BRM	0	1	1954	1960

Mercedes

Again, Mercedes is either 3rd or 4th in most successful.

Parts

Mercedes is probably the team which uses the highest of their own parts.

The only parts they source from outside are the ones mandated by the FIA such as Pirelli tires and McLaren ECU.

Sponsors

Mercedes has their long-term principal name sponsor Petronas. The full official name of the team Mercedes-AMG-Petronas F1 Team, the longest F1 Team name in 2026 season if you go by number of characters.

Famous drivers

The most famous Mercedes drivers are Lewis Hamilton, Micheal Schumacher, Juan Manuel Fangio, and the recent hero Andrea Kimi Antonelli.

History

They first entered F1 as Daimler-Benz-AG in 1954.

The 2 namesakes Carl Benz and Gottlieb Daimler never actually met because the team was a merger of 2 companies and Daimler had died 26 years before the merger.

They quickly rose high with Juan Manuel Fangio.

But on June 11 1955, during a 24 Hour Le Mans Race (an endurance race lasting a whole day and night) a Mercedes-Benz 300 SLR collided with another car and launched into the spectator stands. Over 80 people were killed and over 100 injured. Immediately, the team withdrew all their cars even though they were leading. After this tragic season, they decided to withdraw from all motor racing, including F1.

In 1994, exactly 39 years later, they returned to supply engines for the Sauber F1 team. They forged an alliance with McLaren until 2010. Honda pulled out of F1 after the 2008 season even though they had some nearly finished cars for 2009. Their team boss Ross Brawn bought the team for £1 and rebranded it as Brawn GP. As the cars didn't have engines, they got the engines from Mercedes, which helped them to win both the WDC and WCC. For the 2010 season, Mercedes bought Brawn GP and rebranded as Mercedes GP Petronas. They changed their name to Mercedes-AMG-Petronas in 2012 to promote AMG, the company's high performance division.

Williams Racing

Either the 3rd or 4th most successful team, depending on if you go by race wins or championships

Parts

They design their own chassis, and structural components, but are supplied engines and gearboxes by Mercedes.

They use fuel from Gulf Oil officially but also Petronas which is standard for Mercedes engines.

Sponsors

They are sponsored by Atlassian primarily making their full official name Atlassian Williams Racing.

Famous Drivers

The most famous drivers for Williams Racing are Alan Jones and Ayrton Senna

History

Frank Williams and Patrick Head founded Williams Racing after Williams' earlier unsuccessful racing venture Frank Williams Racing Cars (Later Wolf-Williams racing cars).

Clay Regazzoni won the British Grand Prix in 1979 for Williams Racing.

They won their first WCC and WDC the next year with racer Alan Jones and Carlos Reutmann.

Red Bull

Red bull is the 5th most successful team

Parts

Apart from FIA Regulationed parts, they only outsource internal gears.

Sponsors

Their title sponsor is Oracle, a Cloud Infrastructure and Enterprise Technology company

Famous Drivers

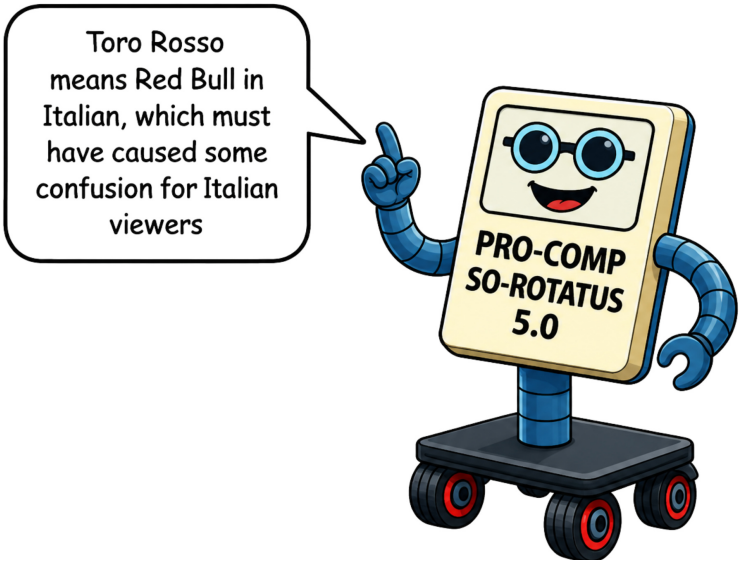
Red Bull has some of the most famous drivers including Max Verstappen and Sebastian Vettel.

History

In 1997, a team called Stewart Grand Prix competed. In 1999, the founder Jackie Stewart sold the team to Ford, who rebranded it as Jaguar racing. Meanwhile an energy drinks company called Red Bull had been sponsoring the F1 Team Sauber since 1995. When Jaguar racing was put up for sale in 2004, Red Bull bought it for only $1. In return, they had a commitment to invest $400 million for at least 3 Grand Prix races.

In their first season in 2005, Red Bull performed massively better than the earlier Jaguar, getting more points than Jaguar had gotten in 2003 and 2004.

They were led by former McLaren driver David Coulthard and the other car was shared by Christian Klien and Vitantonio Liuzzi.



In 2006, they bought the F1 Team Minardi, renamed it as Toro Rosso which housed future Red Bull drivers in it, such as the legendary Max Verstappen. It was renamed for the 2020 season as AlphaTauri, to support Red Bull's fashion brand. In 2024, AlphaTauri was rebranded to Visa Cash App Racing Bulls F1 Team, with 2 primary name sponsors related to money and the longest F1 team name in the 2026 season if you go by number of words.

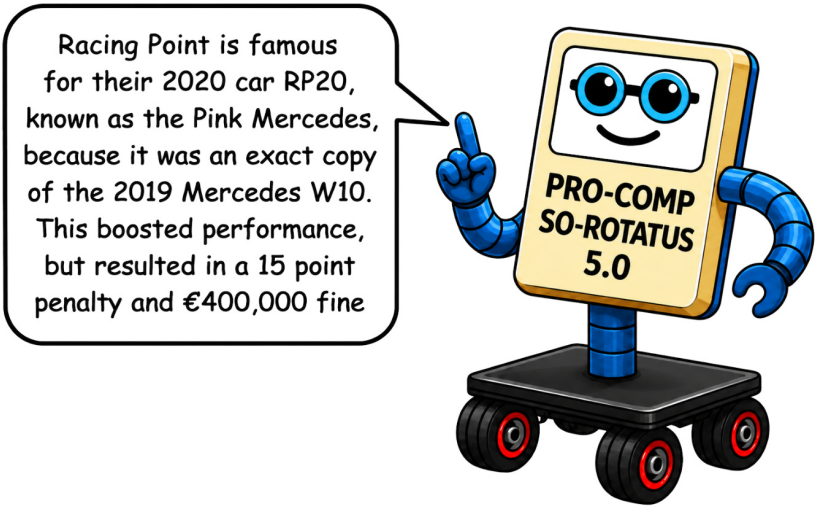
Silverstone and Enstone Lineages

Sometimes, F1 Teams can get rebranded so many times that it's unclear if or when they become different teams.

For example, there was a team called Jordan Grand Prix, founded by Eddie Jordan in 1991, which was the first team Michael Schumacher raced for.

Alex Shnaider bought it in 2006 and rebranded it as Midland F1, which did not perform well, leading it to be sold to sports car company Spyker after a single season. Spyker renamed the team Spyker F1, but they still did poorly.

In 2007, the team was bought by Vijay Mallya, the Indian businessman who bought the IPL (Indian Premier League) team RCB a few months later. The team performed much better in the upcoming seasons. Despite small budgets, they did smart engineering and were regularly in the top 5, though they never won a race. They had drivers like Sergio Pérez, Nico Hülkenberg, Esteban Ocon and defeated teams with much higher budgets.



In 2018, the team entered Administration (similar to Bankruptcy) due to financial problems. Lawrence Stroll bought the team's assets (but not the team itself) and competed as a new team, Racing Point. Racing Point won the 2020 Sakhir Grand Prix, the first win in this lineage since Jordan Grand Prix.

Lawrence Stroll rebranded as Aston Martin in 2021, as he was the Executive Chairman of the Aston Martin Lagonda automobile company.

This whole lineage is called Silverstone lineage, because their factory has been located near Silverstone village since 1991.

There is a similar lineage called the Enstone lineage which goes like this:
Toleman → Benetton → Renault → Lotus Renault GP → Renault → Alpine

FOM and FIA

The FOM (Formula One Management) is a lesser known entity than the FIA (Federation International de l'Automobile) or Formula One Group.

What the FOM really does

• The FOM releases the Formula One rights bundle for broadcasting. This is currently held by Apple TV in the USA and FanCode in India.

• They manage F1 TV, the OTT which shows F1 Races.

• They collect and manage the telemetry data from the cars

• They collect money (Can be upto $60 million!) from the circuit stadiums which host F1 Grand Prixes.

What the FIA does

• The FIA basically established modern F1 Racing.

• They started the Grand Prixes.

• They made the rules and set the regulations.

The FIA used to do everything the FOM does now too, but because of a European Commission ruling in the 1990s, they had to separate their regulatory and commercial actions into 2 separate entities. The FOM was first founded as FOPA (Formula One Promotions and Administration) in 1987. It was renamed to FOM in 1992 and was given the commercial rights in 1995 for 14 years. This was extended to 110 years in 2001.

How the FIA makes money

Obviously, the FIA also has to make money.

When a team enters for a season they have to pay

• Entry fee of $703330 for entering the competition

• Points fee: The reigning champion WCC has to pay $8438 per point they scored in the previous season. The other teams have to pay $7030 per point.

How the F1 Commission Works Together

Together, the FIA, FOM and 10 F1 teams are the F1 commission.

The commission typically calls a meeting around major events like the start and end of a season or when they get a new television partner or new sponsor.

To understand how the commission works, let's look at a hypothetical situation.

All the bodies are assembled, the FIA President (or a representative), FOM executives, one representative for each team.

The FIA President proposes a change: "I propose that neorubber has to be used in all cars for this season and later seasons"

Then comes the discussion.

The teams don't like the idea because they would have to change their design and suppliers. The FIA puts a counter argument that these tires are safer and less likely to melt. The FOM might not like that it is a very simple material, not being very interesting for audiences, but would try to negotiate instead of publicly rejecting it because people would think they are ignoring safety. The FIA points out that it is the cutting-edge, ultramodern material of the year. The discussion is usually the longest part of a meeting.

Then the official vote takes place. The FIA, Formula One Management (FOM), and all the Formula 1 teams vote on the proposal. In our example, the FIA, FOM, and three teams support the idea, while the other seven teams oppose it. Because major rule changes usually require more than a simple majority to be approved, the proposal does not receive enough support and is rejected.

Jokes

1. In most sports, winning gives you money. In Formula 1, winning also gives you a tighter budget.

2. In Formula 1, teams are constructors, because they construct cars but destroy bank accounts.

3. Some F1 Team lineages are so long they go back to the Stone lineage.

Quiz

1. In which year did the 'Pink Mercedes' race? _____

2. What was the FOM first called?

_____

3. In which year did McLaren first use their iconic papaya orange color? _____

4. What is Ferrari's full official name as of the 2026 season? _____

5. When did the Mercedes Benz racing team first enter F1 and under what official name?

_____,_____

6. Which team has the longest official name as of 2026 (write the full official name)

_____

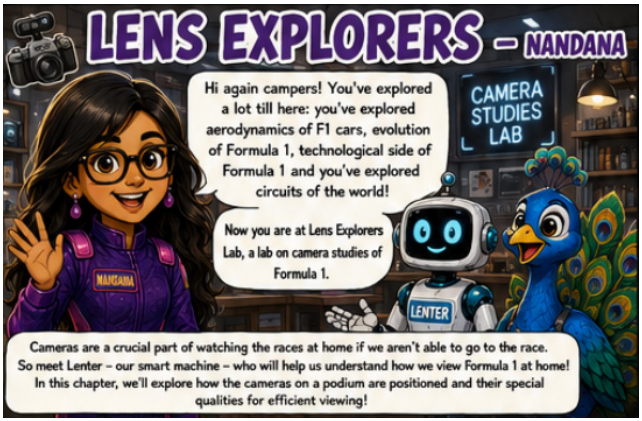
7. Which team made a 6 wheeled F1 car and what was its model? _____

8. Which team's parent company makes an energy drink?

_____

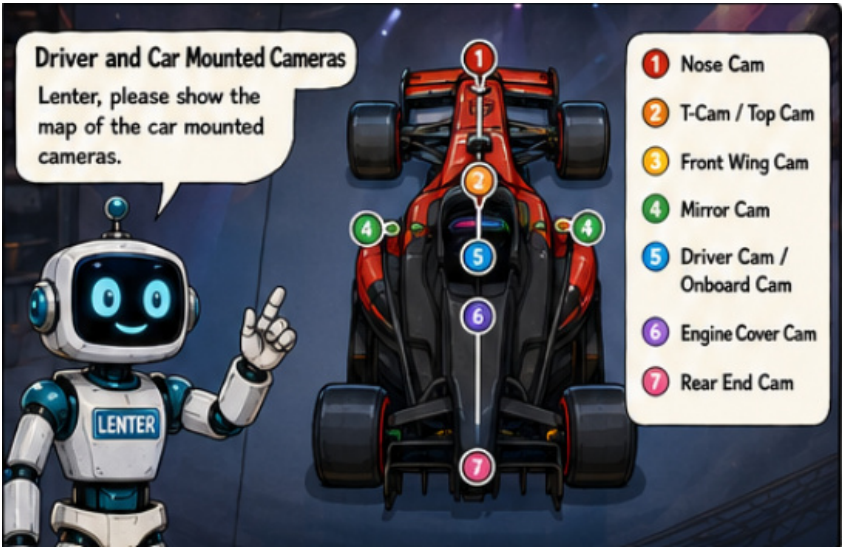
CHAPTER 08

LENS EXPLORERS



Hi again campers ! You've explored a lot till here : you've explored aerodynamics of F1 cars , evolution of Formula 1 , technological side of Formula 1 , teams, profit and you've explored circuits of the world! Now you are at Lens Explorers lab , a lab on camera studies of Formula 1 . Cameras are a crucial part of watching the races at home if we aren't able to go to the race. So in this chapter we have a machine known as Lenter . "Lenter" will help us understand how we view Formula 1 at home? So in this chapter we will explore the way the cameras of a podium are positioned and their special qualities for efficient viewing!

Driver and Car Mounted Cameras



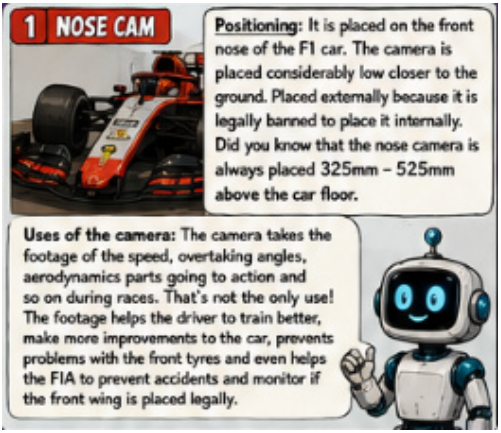
Lenter, please show the map of the car mounted cameras.

To record close - ups of the car there are often cameras mounted on the car . So let's explore the different lenses mounted on the car:

1. Nose Cam :

a. Positioning : It is placed on the front nose of the F1 car. The camera is placed considerably low closer to the ground. Placed externally because it is legally banned to place it internally.

Did you know that the nose camera is always placed 325mm - 525mm above the car floor.



b. Uses of the camera : The camera takes the footage of the speed , overtaking angles , aerodynamics parts going to action and so on during races . That 's not the only use of the footage helps the driver to train better, make more improvements to the car , prevents problems with the front tires and even helps the FIA to prevent accidents and monitor if the front wing is placed legally.

2. T- Cam / Top cam :

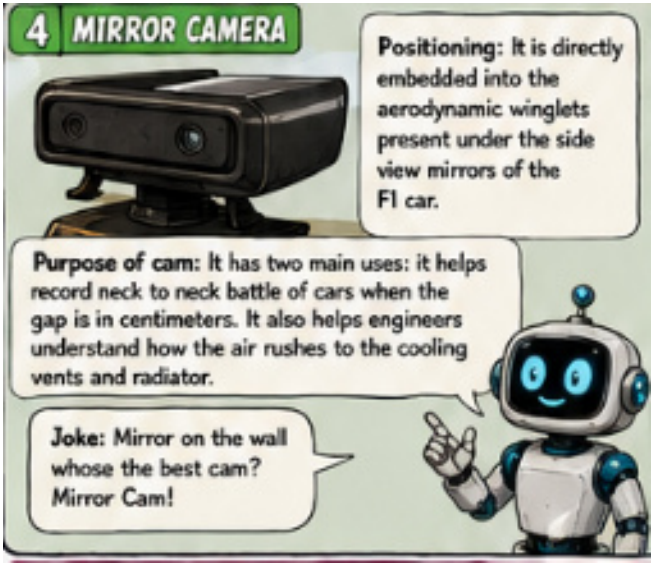
a. Positioning : It is positioned at the Mt. Everest (highest point) of the car . It is placed above the driver 's head on top of the roll loop air intake (a combustion and safety scoop)

b. Purpose of this camera : The camera is used for broadcast footage recording (it gives front and rear view footage providing effects that makes viewers enjoy it), telemetry (it has data antennae and GPS that help track location and report data to the FIA.). The last is super interesting, it is painted in yellow or black to differentiate the drivers from the same team !

3. Front wing cam :

a. Positioning : As you learnt in the first chapter there are two wing plates on either side of the nose that help in the aerodynamics. So these cameras are embedded right on it .

b. Purpose of camera :So this camera is Amulya 's favourite because it gives viewers an amazing sense of the speed of the car. Not only it helps us view the structural physics : Pirelli tires flex and blister when the speed is this intense and it shows how the load suspension works live! It even helps understand if the driver has gotten into a wing plate accident easily (sheesh!) .



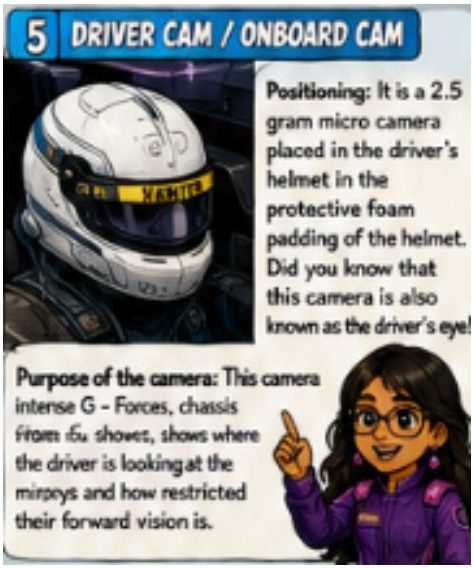
4. Mirror Camera :

a. Positioning: It is directly embedded into the aerodynamic winglet's present under the side view mirrors of the F1 car . (Joke : Mirror on the wall, who's the best cam ? Mirror Cam)

b. Purpose of cam: It has two main uses : it helps record neck to neck battle of cars when the gap is in centimeters. It also helps engineers understand how the air rushes to the cooling vents and radiator.

5. Driver Cam/ Onboard Cam:

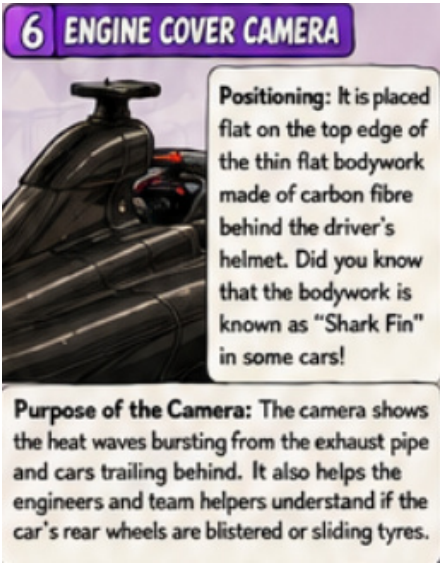
a. Positioning : It is a 2.5 gram micro camera placed in the driver's helmet in the protective foam padding of the helmet. Did you know that this camera is also known as the driver's eye!



b. Purpose of the camera: This camera records the intense G - Forces , chassis vibrations , shows where the driver is looking at the mirrors and data displays and how restricted their forward vision is .

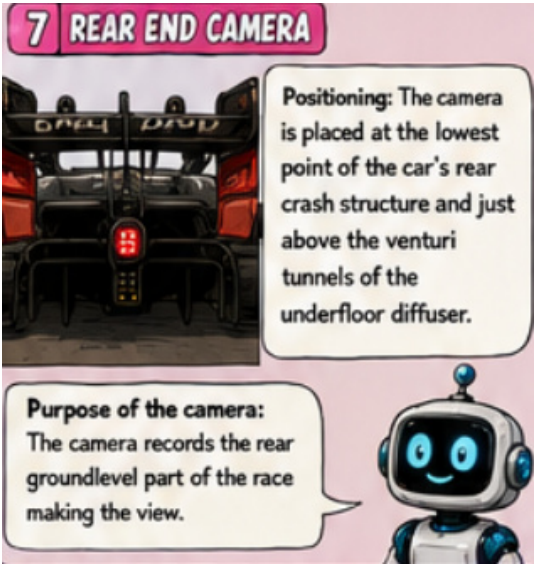
6. Engine Cover Camera:

a. Positioning : It is placed flat on the top edge of the thin flat bodywork made of carbon fibre behind the driver's helmet. Did you know that the bodywork is known as "Shark Fin" in some cars!



b. Purpose of the Camera: The camera shows the heat waves bursting from the exhaust pipe and cars trailing behind. It also helps the engineers and team helpers understand if the car's rear wheels are blistered or sliding tires.

7. Rear end Camera :



a. Positioning : The camera is placed at the lowest point of the car's rear crash structure and just above the venturi tunnels of the underfloor diffuser.

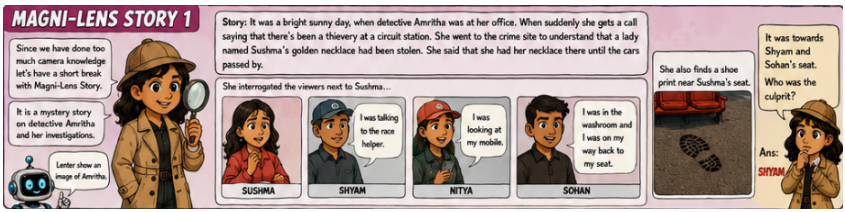
b. Purpose of the camera : The camera records the rear groundlevel part of the race making the view.

Magni-Lens Story 1

Since we have done too much camera knowledge let's have a short break with Magni - Lens Story. Before we start this let me tell you what a Magni- Lens Story is . It is a mystery story on detective Amrita and her investigations. I will give you clues so you can decide on the culprit .

Lenter show an image of Amrita;

Story: It was a bright sunny day, when detective Amrita was at her office . When suddenly she gets a call saying that there's been a thievery at a circuit station . She went to the crime site to understand that a lady named Sushma 's golden necklace had been stolen . She said that she had her necklace there until the cars passed by. She interrogated the viewers next to Sushma, Shyam, Sohan and Nitya . Shyam said, "I was talking to the race helper ." Nitya said , "I was looking at my mobile. " Sohan said, "I was in the washroom and I was on my way back to my seat. She also finds a shoe print near Sushma 's seat. It was towards Shyam and Sohan 's seat. Who was the culprit ?



Ans

—

Okay the answer is (*urshyS*)

Ground-Level and Track-Embedded Cameras

Till now we have explored cameras on the car and the driver . But that's not it, there are many more cameras. There are cameras placed on the track and are placed pretty low . Lenter, Amulya and I are going to take you through some of these. But before that I have to tell Amulya the way to call her friends at her hometown, the Buddh International circuit (that's where we met) . And there we go I am back so where were we? Ah! Ground Level and Track embedded cameras ok then let's start.

1. Start/Finish Camera:

a. Positioning : These cameras are placed at overhead start-light gantry . The cameras are placed there looking straight down at the start line

b. Purpose of the camera: It helps understand who won when the competition is so neck and neck . It even helps to track how many laps are made by each car .

2. Apex Camera

a. Positioning : The camera is placed low to the ground and is placed exactly at the geometric midpoint of the corner.

b. Purpose of the camera: The camera is used for seeing track limit violations and cornering speed.

3. Kerb Camera

a. Positioning :It is placed inside the concrete rumble strips (ridged blocks placed at the edge of the track).

b. Purpose of the camera: Records low and slow motion suspension loads .

4. Braking Zone Camera:

a. Positioning: It is placed upon high towers or elevated safety points during a F1 race.

b. Purpose of the camera: Capture accidents ,tire locking and thrilling overtakes .

5. Pit Entry Camera:

a.

Positioning:It is placed right at the demarcation line before the pit entry.

b. Purpose of the camera:The camera is used to see if drivers slow down at the pit speed limit line and illegal crossings at the pit entry blend.

6. Pit Exit Camera:

a. Positioning: It is positioned at the end of the pit lane looking down at the acceleration blend lane .

b. Purpose of the camera: Ensure safe join back of the driver onto the main track.

Magni Lens Story 2

So let's take a short break with Magni Lens Story again. But this time I am not the narrator Amulya and Lenter are! Amulya and Lenter, " Yes let's start:"

Story: It was one normal Sunday and Amrita was at home relaxing . When she gets a call saying that the water at the place had contaminated not as an accident but on purpose because of the water system being broken by someone . She first started by interrogating the head of the circuit when the water system was broken . He said it was last Monday at 5pm (working hours) when it was working just fine. So she checked the CCTV cameras. There was one person in the camera who opened the system . He wore a black jacket . There were three men who had a black jacket . Ram, Suresh and Akash , she interrogated them.

Ram said ," I was at home sleeping." Suresh said," I was working at my table ." Akash said, "I was in the washroom here." Who was the culprit?

Ans_____

The answer is: (𝓌𝑒𝓋)

Cameras In The Sky

We have explored a lot till now . The cameras which we explored were towards the ground but now let 's explore the Sky cameras.

1. Helicopter Camera:

a. Positioning : It is placed on the nose of the helicopter or the side. Which flies 300-1000m above the circuit.

b. Purpose of the camera : The camera shows the field and grid view of the circuit . It even shows the gap between leading and other cars and the entire circuit.

2. Drone Camera:

a. Positioning : Mounted on a cinema drone, used only where allowed and kept within designated airspace during selected parts of the event.

b. Purpose of the camera : It is used to record close ups of individual cars . It even captures complex corner sequences.

3. Cable Cam:

a.

Positioning : suspended on a high tension network of steel cables .

b. Purpose of the camera : It goes at the same speed of the car to take side by side tracking of the car.

4. Crane Camera:

a. Positioning : placed on a motorized mechanical arm fixed to the tripod.

b. Purpose of the camera : The camera takes from close up to upward smoothly up to 30 ft.

The cameras have helped Amulya, Lenter, and I see F1 through a different lens. A different perspective to the effects we see at home . Just a lens takes us to the circuit far away in another country . We just see the race , the drivers and cars , but that's not the only part there is another part of the lens . Next time you see a Formula 1 race remember these heroes that help you see the thrilling race.

Jokes

Lenses need not just be a studies thing, it can even be humorous . Let me even show you the funny side:

1. Which flying camera flies in high tension?

Ans : The Cable cam of course.

2. Why was the Braking Cam crying ?

Ans Because it captured too many accidents.

3. Why did the Apex camera like geometry?

Ans : Because it was placed at the geometric midpoint of the corner.

4. Which flying cam likes the cinema?

Ans : The Drone cam of course

Quiz

So coming to the end let's end it with a short fun Quiz:

1. What is your favourite Camera?

2. Which camera is used for seeing track limit violations and cornering speed?

3. Which camera is suspended on a high tension network of wires?

4. Which camera is placed on a motorized mechanical arm?

5. Which camera is placed on the demarcation line before the pit entry?

CHAPTER 09

SPEEDWAY STARS

Hi there, I am Shaurya, an F1 driver in the Formula One Racing Camp, a great admirer of the stars of the circuit… Now you are soon coming to an end to the 1st Formula Racing Camp but don't forget THE HALL OF FAME of the SPEEDWAY STARS! Till now you saw the structure, past and even the future. What is a book without drivers! We will now talk about the DRIVERS OF F1! So are you ready to dive in?

Giuseppe "Nino" Farina

He was probably the first Formula 1 champion 🏁 Yaay! He was born in Turin, Italy in the early 20th century damn that's so back. A city famed for car-buildings, coach buildings and fast machines. His father, Giovanni Carlo Farina, had founded the coach-works **Stabilimenti Farina**, and the racing world would prove a natural playground for young Nino. Though he earned a doctorate in political-science (or engineering, depending on source) and began a promising academic path, racing called him .His first competitive outing was the 1925 Aosta–Gran San Bernardo hillclimb, where a crash ended his run prematurely — the same determination that led to crashes would also define his career. With the world of racing evolving rapidly, Farina made his way from hillclimbs to Grand Prix cars throughout , aligning himself with teams like Maserati and Alfa Romeo, and earning the mentorship of Italian legend Tazio Nuvolari. He won the inaugural 1950 season, driving for **Alfa Romeo.**

Juan Manuel Fangio

Juan Manuel Fangio was born on June 24, 1911, Balcarce, Argentina. He was a legendary F1 driver.

Known affectionately by nicknames such as "**El Chueco**" (the bowlegged one) and "**El Maestro**" ('The Master' or 'The Teacher'), Fangio was a founding father of Formula 1, defining the sport's earliest seasons. Fangio clinched the World Drivers' Championship five times during his F1 career, an achievement that remained unbeaten for nearly five decades and at a time when the sport was most dangerous. Fangio's journey into motor racing began from humble origins. With a youthful dislike for anything academic, he preferred a more practical vocation in auto mechanics. His passion for cars soon propelled him onto the racing scene, where he made his debut in the 1938 Turismo Carretera series, a popular stock car racing series in Argentina, driving a Ford V8 (crazy we know).

By 1940, now with **Chevrolet**, he won the championship then and also again the following season.

With **Mercedes-Benz** in 1954 and 1955, and **Ferrari** in 1956.

One of the Oldest F1 Records That Remain Unbroken.

In addition to five Drivers' titles, Fangio left the sport with a staggering winning percentage of 46.15%, 24 victories out of 52 races, Fangio's 1957 German Grand Prix victory at 46 years, 41 days old is the stuff of legend. The third oldest F1 driver to win a race, after a disastrous pit stop dropped him far behind the leaders, he drove arguably the most inspired race in F1 history, overtaking both Ferrari drivers with just a couple of laps to go.

Alberto Ascari

Alberto Ascari was born on 13 July 1918 in Milan, Italy. He was the son of Antonio Ascari, a prominent Grand Prix motor racing (the sport which would eventually become F1) driver for Alfa Romeo who tragically died in a crash during the 1925 French Grand Prix. Alberto remained drawn to motorsport from an early age, though the loss deeply affected him.

At just 19 years old, he began his professional racing journey on two wheels, signing with the Bianchi motorcycle team. However, his transition to car racing came shortly after. In 1940, Ascari entered the Mille Miglia, one of Italy's most prestigious road races.

He drove an Auto Avio Costruzioni 815, a car developed by Enzo Ferrari before the official founding of Ferrari.

During World War II, rather than serving in a military role, Ascari ran a fuel transport business for the Italian army alongside fellow racer Luigi Villoresi, supplying fuel to army depots in North Africa.

How did Ascari's relationship with Ferrari really begin?

After World War II, Ascari resumed his career, initially racing for Maserati in 1947 and 1948 alongside his close friend Luigi Villoresi. Their strong performances caught Enzo Ferrari's attention. By 1949, Ascari officially joined Scuderia Ferrari, which had just begun building its own cars for Formula 1 and other sports car competitions. Ascari quickly delivered results. In 1949, he secured five major victories driving for Ferrari: Bremgarten in Switzerland, Monza in Italy, Palermo in Argentina, Silverstone in England and Palermo in Argentina once more. Ascari's calm, analytical driving style and technical feedback made him an ideal fit for Ferrari's racing program. He quickly became one of the team's leading drivers during its formative years in Formula 1. Alberto Ascari driving the Ferrari 125 F1 at the 1950 Monaco Grand Prix

Alberto Ascari made an early mark in Formula 1 when he finished second at the 1950 Monaco Grand Prix, becoming the youngest driver at the time to score both championship points and a podium, at 31 years and 312 days old. His performances during the early seasons of Formula 1 helped establish Scuderia Ferrari as a competitive stable. Notably, he played a key role in the development of Ferrari's early Grand Prix cars, particularly the Ferrari 375 F1.

In 1952, the Formula 1 switched to Formula 2 regulations, limiting engines to 2.0 litres. Ferrari introduced the Ferrari 500, which Ascari drove to great success. He missed the season opener (the Swiss Grand Prix) while qualifying for the 1952 Indianapolis 500, a race that was part of the Formula 1 Championship calendar at the time. After returning to Europe, Ascari dominated the season by winning the remaining six rounds, setting the fastest lap in each, and also securing five additional victories in non-championship events. This performance earned him his first Formula 1 Championship title.

Alberto Ascari at the 1950 Monaco Grand Prix

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Notably, he played a key role in the development of Ferrari's early Grand Prix cars, particularly the Ferrari 375 F1. During his Formula 1 career from 1950 to 1955, Alberto Ascari competed for Ferrari, Maserati, and later Lancia. He started 33 World Championship Grands Prix, winning 13 races, securing 17 podiums, 14 pole positions, and setting 12 fastest laps. His back-to-back Formula 1 Championships in 1952 and 1953 made him the dominant driver of his era and Ferrari's first Formula 1 Champion. Ascari's record of seven consecutive Grand Prix victories (set across the 1952 and 1953 seasons) remained unmatched for decades. Ascari's success marked Ferrari's first major breakthrough in Formula 1 and helped establish the team's long-term presence in the sport. Ascari died on 26 May 1955, at age 36, after crashing during a test at Monza in the Curva del Vialone corner. The incident eerily mirrored his father's fate: both died at the age of 36, both in racing crashes, and both near Milan. His name lives on in Formula 1 history, and the Ascari chicane at Monza is named in his honour.

Mike Hawthorn

Mike Hawthorn loved life, drove fast and died young. Big, blond and boisterous, he often raced wearing a broad grin and a bow tie. He regarded motorsport as a quick way to further the fun he constantly pursued.

When his pastime became a profession he partied as hard as he drove, though his career was also tinged by tragedy, scandal and personal misfortune. Near the end he found Formula 1 racing no fun at all, but he went out a winner. Other champions were greater drivers but none was a more colourful personality.

Had he been born a decade earlier, John Michael Hawthorn might have been a heroic Spitfire pilot in the Battle of Britain.

Instead, the circumstances in his formative years led him into motorsport and he became **Britain's first** world driving champion. Two years after Mike's birth on April 10, 1929, his racing enthusiast father Leslie bought a garage near the Brooklands circuit in Farnham, Surrey, where he had raced motorcycles prior to World War II. Inspired by the proximity of Brooklands, and by the atmosphere in the Hawthorn garage where cars and motorcycles were prepared for competition, Mike was only nine years old when he decided he wanted to become a racer. His education, at a prominent Public School followed by studies at Chelsea technical college and an apprenticeship with a commercial vehicle manufacturer, was intended to prepare him for a career at the Farnham garage. Meanwhile, his father also encouraged Mike's interest in motorsport, providing him with motorcycles, then cars for local competitions. In 1950 he began winning races in a small Riley sportscar bought for him by his father. Three years later the **'Farnham Flyer'** was driving a Formula 1 car for Enzo Ferrari.

Mike's meteoric rise from club racer to Grand Prix driver took place within one momentous afternoon at the 1952 Easter Meeting at the Goodwood circuit. It was his first competition in a single-seater, a Formula 2 Cooper-Bristol provided for him by a family friend, and the opposition included the famous Argentine drivers Juan Manuel Fangio and Froilan Gonzalez.

Mike won the F2 race from pole position, then also finished first in the Formula Libre race and was a sensational second in the main event, for Formula 1 cars.

Impressive results aside, the Farnham Flyer was a commanding figure in the spindly little Cooper, with the top half of his 6 foot 2 inch (188 centimetre) frame towering above the cramped cockpit, elbows flailing in the wind, head thrust forward, chin first. Prior to Goodwood he had always raced in his everyday clothes, usually a sports jacket and a tie, which at speed tended to flap in his face. For his single-seater debut Mike bought white overalls and wore the bow tie that became his trademark.

Inspired by his splendid showing at Goodwood, Mike and his father decided to enter the Cooper in the remaining races of a 1952 Formula 1 season that was being dominated by Alberto Ascari in a Ferrari. A fourth place in Belgium, a third in Britain and another fourth in Holland left Mike an astonishing fourth overall in the standings. Enzo Ferrari was impressed and hired him for 1953.His education, at a prominent Public School followed by studies at Chelsea technical college and an apprenticeship with a commercial vehicle manufacturer, was intended to prepare him for a career at the Farnham garage. Meanwhile, his father also encouraged Mike's interest in motorsport, providing him with motorcycles, then cars for local competitions. In 1950 he began winning races in a small Riley sportscar bought for him by his father.

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His only championship victory in his first Formula 1 season was a singular feat that became the highlight of Mike's career. In a classic 1953 French Grand Prix at Reims his Ferrari crossed the finish line a hair's breadth ahead of the Maserati driven by the great Fangio. On the podium, when he heard ‘God Save The Queen’ being played Mike burst into tears and was warmly embraced by the ever-gallant Fangio, who thought him: "a nice young fellow, always in a good mood."

To the French, the Englishman who raced with a bow tie became known as 'Le Papillon' (The Butterfly). At home, Britain's flamboyant new racing hero was subjected to close scrutiny and flaws were sought by sensation-seeking tabloid newspapers. He was accused of evading compulsory military service, though in fact, he had been rejected because of a chronic kidney ailment. But his reputation suffered, and fate also conspired against him. Early in 1954 his arms and legs were badly burned in a crash in a non-championship race in Sicily. Then his father was killed in a road accident.

first accused of triggering the accident and, though he was later exonerated, the deadly side of motorsport badly bothered him.

Hawthorn celebrated his first Grand Prix victory at the 1953 French Grand Prix.

His racing life improved in 1957 when he returned to Ferrari, where he also found a kindred spirit in a fun-loving team mate. Mike's championship year of 1958 was blighted at Germany's notorious Nurburgring where he saw his team mate's Ferrari Dino 246 crash with fatal results. The death of Collins left him devastated and disillusioned and Mike only reluctantly completed the season. He finished one point ahead of Vanwall's Stirling Moss (who had won four races to Mike's single victory, in France) to become the first British World Champion.

By now engaged to Jean Howarth, a beautiful fashion model, Mike had lost the heart for racing and at the end of the year he announced his retirement. But he continued to drive fast on public roads and on January 22, 1959, near his Farnham home, his Jaguar skidded off a wet corner and 29-year-old Mike Hawthorn was killed.

Jack Brabham

Hello there, campers I can't believe you have read till here!

So the image to the right is Jack Brabham, he was one of the coolest drivers of that time. Jack won 3 world championships that were a product of both his engineering experience and his amazing driving skills. His first two titles, in rear-engined Coopers he helped develop, confirmed the obsolescence of front-engined Formula 1 cars. It was really hard to race in those cars at that age like you saw in the previous chapters.

Other Famous Drivers

Phil Hill

Mr. Phil Hill was the first American driver to win the world championship. He was born Miami, Florida on April 20th 1927

John Surtees

Famed as the only World Champion on both two and four wheels, John Surtees rode motorcycles and drove Formula 1 cars with equal skill and bravery.

Denny Hulme

Denny Hulme hated fame and being popular. He was also known as “bear”. He was a man who could not express his feelings. He was an unspectacular race driver; he shortly died in a race behind the wheel.

Jackie Stewart

His outstanding track record still ranks him among the most successful champions. But Jackie Stewart is also known for influencing the rules and safety of the sport. His one-man safety crusade made the sport much safer. His excellent communication skills helped make it more popular. He set new standards of professionalism for drivers and was also a pioneer in exploiting Formula 1 racing's commercial potential. His keen intelligence and tireless energy helped, but he would never have been able to exert such influence had he not been a truly great driver.

Mario Andretti

Mario Andretti personified the American Dream, an immigrant who came to the land of the free and the home of the brave and achieved considerable fame and fortune.

He did so in a job he would have done in anonymity and for nothing. Few drivers loved their racing more, and few suited stardom as much as the congenial man whose unforced charm was as natural as the talent that brought him so much success in so many types of racing. His insistence on racing in his adopted homeland meant he came late to realising his boyhood ambition of becoming Formula 1 World Champion.

Jody Scheckter

He exploded on the scene as an erratic, crash-prone wild man whose desperate deeds of derring-do put himself and his peers in great danger. Jody Scheckter became infamous for causing one of the biggest accidents in Formula 1 history, after which there were demands that he should be banned from the sport. Instead, he straightened himself out and concentrated his considerable talent and ambition on becoming World Champion. Having achieved his goal (with Ferrari, whose next champion would only be 21 years after him), he quickly retired.

Alan Jones

A straight-talking, iron-willed, hard-driving tough guy, Alan Jones fought his way to the forefront, where he defended his territory with ruthless determination and large doses of intimidation. His belligerence was partly a by-product of a long and hard struggle to make it to Formula 1 racing in the first place.

Once there, he was considered little more than a journeyman driver, until he teamed up with the then equally undistinguished Williams team. Together, they took on the world and beat it, with AJ becoming the prototypical Williams driver.

Nelson Piquet

He was never a dominant driver but a crafty expert in winning by stealth, according to his detractors. He admitted he was lazy, yet willingly worked hard to improve his car. He could be cold and cruel but could also be warm and funny. He hated being a celebrity yet lived the life of a playboy to the hilt. He was never hugely popular but couldn't care less. All Nelson Piquet really cared about was driving a racing car, which he loved with a passion, especially when he won, which he did often enough to become a triple World Champion.

Keke Rosberg

The original Flying Finn was a swaggering swashbuckler whose dashing style of driving enlivened every race he was in. He was a late arrival in Formula 1 racing, having wheeled and dealed and raced his way around the world in other categories for a dozen years, but tried harder than ever to make up for lost time. In the record books his name is not near the top in terms of Grand Prix victories, yet in that select category of those who actually looked as fast as they drove the wonderfully aggressive Keke Rosberg ranks among the very highest.

Alain Prost

His place in Formula 1 history as one of the sport's greatest drivers is secure, though a career full of conflict and controversy detracted somewhat from his considerable achievements. He won four championships but also left teams acrimoniously on four occasions. He made winning races - 51 times - look easy but was less successful at the politics in which he was invariably embroiled. His bitter feud with Ayrton Senna brought out the best and worst in them both. And yet among the champions only Lewis Hamilton, Michael Schumacher and Juan Manuel Fangio won more crowns than Alain Prost.

Ayrton Senna

People, you might know him as a legend. I call him a legend. He was the kindest person but after all this he still had a tragic death. No one tried harder or pushed himself further, nor did anyone shed so much light on the extremes to which only the greatest drivers go. Intensely introspective and passionate in the extreme, Ayrton Senna endlessly sought to extend his limits, to go faster than himself, a quest that ultimately made him a martyr but did not diminish his mystique. For those of you who do not know him, he crashed into a concrete barrier in 1994. Thankfully, such an incident cannot happen in modern times.

Michael Schumacher

He is one more legend after Senna. Formula 1 World Drivers' Championship began in 1950 the title has been won by 34 different drivers, 16 of whom won more than one championship. Of the multiple champions the most prolific was Juan Manuel Fangio, whose record of five titles stood for five decades until it was eclipsed by the most successful driver in the sport's history. Seven times a champion, Michael Schumacher also holds nearly every scoring record in the book by a considerable margin. Though his ethics were sometimes questioned, as was his decision to make a comeback after retiring, his sheer dominance when in his prime is beyond doubt...

Fernando Alonso

The 28th Formula 1 World Drivers' Champion was at the time the youngest ever. Just 24 years old, Fernando Alonso also led the Renault team to the 2005 Constructors' Championship, thus ending the reign of the Michael Schumacher-Ferrari combination that had dominated for so long. The precocious and personable youngster who made so much history so soon comfortably wore the crown - a bright, polished, perfectly poised new star. Confirmation of his brilliance came in 2006 when he successfully defended his title against strong opposition from Schumacher, whose subsequent retirement left Alonso well-placed to succeed him as Formula 1's resident superstar...

Kimi Raikkonen

Fast-tracked into the sport with the shortest CV on four wheels, the unknown newcomer who came from nowhere and said next to nothing immediately proved he knew exactly what he was doing: driving a Formula 1 car as fast as it could possibly go. The car couldn't always keep up with his talent and it took seven seasons for Kimi 'Iceman' Raikkonen to become World Champion. Notoriously inanimate and uncommunicative, the silent speedster's frozen expression in fact masked the hidden depths in one of the most original.

Jokes

Why did the Red Bull car go to the gym?

To improve its downforce. You get it?

What was Ferrari's race plan:

1: Make a strategy. Ok understandable

2: Change the strategy. Last moment change

3: Forget the strategy. And win!

It's just a joke, we're also Ferrari fans, especially Charles Leclerc is the G.O.A.T and also Max Verstappen and Sir Lewis Hamilton on the top.

So guess who's next…… SIR LEWIS HAMILTON!

Lewis Hamilton

Lewis Hamilton is one of Formula 1's most successful drivers taking his first title in the most dramatic fashion - on the last corner of the last lap of the last race of one of the most scintillating seasons. In only his second year at the pinnacle of motorsport, was a welcome human interest story that focused unprecedented international attention on Formula 1 racing. His prodigious talent and pleasing personality made Lewis Hamilton an inspirational role model and ideal ambassador for his profession...Did u know that lewis hamilton got an award (a knighthood) that made him so called SIR. Recently he won the Barcelona GP and that was his P1 in a Ferrari.let's go…

Jenson Button

The 2009 World Champion grew up in the sport, maturing from a carefree youngster distracted by having too much too soon into a seasoned survivor who toiled in obscurity for a decade before his perseverance suddenly paid off in a fairy tale finish with a team that came from nowhere. His unexpected title triumph was partly the product of circumstances that prevented other contenders from overcoming his sensational head start, but ultimately it was Jenson Button's talent and tenacity that took him to the top and got him a pretty good win.

Sebastian Vettel

At the age of 23, he provided a surprise ending to an intensely competitive 2010 campaign. There was surely more to come from this fast rising racer, and it came quickly when he completely dominated in 2011 to become the youngest double world champion. In 2012, with a series of comebacks from setbacks against the strongest opposition in the longest ever season, his third driving title in a row was hard-earned and well-deserved. In 2013, and still just 26 years old, he made short work of becoming the youngest ever quadruple world champion. In truth the precocious youngster who came so far so fast was in a hurry from the time he was a toddler...

Max Verstappen

Everyone tutututu Max Verstappen tutututu. Okay know you know he is my favourite driver so hope u like him.

he's a great driver one of my favourite drivers at number 3 so are u ready to learn more be ready.In 2021 24-year-old Max Verstappen became the 34th Formula 1 World Champion, taking the title from seven-time champion Lewis Hamilton amid controversy on the last lap of the last race. Driving on the limit – sometimes over it – their intense rivalry became a personal feud that pushed them far beyond the rest of the field. The enthralling battle between the ultra-confident Dutch youngster and the most successful driver in the sport's history captured the imagination of a worldwide audience as never before.

Lando Norris

On the final day of the 2025 Formula 1 season, Lando Norris became the sport's 35th World Champion, surviving a brutal test of nerves in the first three-way title decider for 15 years. This was a season for comebacks: first Norris, clawing back a 34-point deficit to McLaren team mate Oscar Piastri, and then, more improbably, reigning champion Max Verstappen making an end-of-season charge that McLaren boss Zak Brown likened to a horror movie character that just keeps coming back – but Norris held his nerve, vanquished the monster… and nicely set-up a string of sequels.

Lando Norris was born in Bristol, in the final weeks of 1999, and raised in the nearby town of Glastonbury. He had an early fascination with motorcycles – venerating Valentino Rossi – but followed Oliver, his older brother, into karting.

CHAPTER 10

FORMULA=1!- JASKIRAT

Hello again, future champions, Jaskirat here — professional F1 driver, owner of the Neon Raptor Vortex GT-X, and somehow still mentally stable after the day yesterday of working with Igneous ('Cause he's SUPER excited!). So, meanwhile I'm at the airport in Monaco with my new Froster Raptor Vortex GT-X (Still not 100% complete). Why did you ask?- The cool 'n ice comic strip on top would have helped you out a little! Yes, that's right. He has been on this adventurous and long journey to find what we needed. We needed a person who could calculate before the thing even happened (That is not even mortally possible).

Someone who can calculate faster than Pro-Comp-So-Rotatus and Nithilan the Formula Racing Camp coach. As the new judge who is biased toward Crimson Cow always says they won even when they lose! So we need someone who can calculate properly and prove that we, Formula Racing Camp, won the race that time, so we can get the trophy back. Igloo would also help us in future races. So that's why we need Igloo! Oh, they're here!!

Jaskirat: "FINALLY! Took you two long enough. Did Igneous stop every five minutes to stare at shiny rocks like I thought?"

Igneous: "THEY WERE ANCIENT CRYSTALS!! One of them looked like my face!"

Igloo: "He tried bringing three of them through airport security."

Igneous: "They called them 'dangerous magical objects. Very rude."

Jaskirat: "Yeah… because glowing frosted crystals which are 9 feet tall usually scare normal people."

Igloo: "Your world is quite strange."

Jaskirat: "You haven't even seen the Formula Racing Camp cafeteria yet."

Igneous: "One time we had a cook contest where we had to cook on an F1 car."

Jaskirat: "Yeah it was kinda weird, but we won though!"

Igloo: "...I already regret leaving my kingdom."

Igneous: "Too late now! Besides, Formula Racing Camp needs you. The Crimson Cow judge keeps giving us second place even when we literally cross first!"

Jaskirat: "Last race we won by 4.2 seconds and he still said, 'Hmm yes… Crimson Cow won by 10 seconds!'"

Igloo: "That is not how mathematics works."

Jaskirat: "EXACTLY! That's why you're here. We need someone whose calculations are sharper than race strategy software."

Igloo: "Then I'll calculate everything myself!"

Igneous: "OH NO."

Jaskirat: "What do you mean 'oh no'?"

Igneous: "Last time she calculated something, she predicted a snowstorm THREE DAYS EARLY."

Igloo: "And I was correct."

Igneous: "Yeah but the kingdom got buried in 40 feet of snow."

Jaskirat: "You know what? You'll fit perfectly into Formula Racing Camp."

Igneous💬: "I'm really tired, I'm gonna go to bed…"

Igloo💬: "Ok, let get this chapter STARTED!!!"

Why Does Mathematics Actually Matters in F1?

Now that Igloo has officially survived the airport, security guards, and Igneous' "totally safe magical crystals," it's finally time to begin the real chapter — mathematics.

And before you say:

"Ughhh… math?"

Trust me. Even Formula 1 drivers say that sometimes.

But here's the thing:

F1 cars may look like giant rockets with wheels, but behind every overtake, every pit stop, every victory, and every championship… There's mathematics hiding somewhere in the background like a sneaky engineer with 14 computer screens.

Without math:

Cars wouldn't know how much fuel to use

Teams couldn't calculate tire wear

Drivers wouldn't know braking distances

Aerodynamics wouldn't work properly

And Igneous would probably try measuring things using "vibes."

Igloo already looked disappointed after hearing that.

Igloo: "Please tell me your team does not actually calculate things using vibes."

Jaskirat: "Only on Wednesdays when Igneous is there."

Igloo: "…That explains many things."

See? Mathematics is everywhere in racing.

When drivers turn corners at insane speeds, teams calculate the perfect angle, speed, grip level, tire temperature, fuel load, lap timing, and even the probability of rain.

Some teams even simulate entire races thousands of times using computers before the real race even begins. Imagine playing the same race over and over again until you discover the perfect strategy.

That's basically what modern F1 teams do.

And honestly?...

That's kind of awesome.

Even your everyday life secretly uses mathematics:

Your phone battery percentage

Video game physics

Pizza slices

Wi-Fi signals

Music rhythm

Shopping discounts

—all involve numbers and calculations.

Mathematics is basically the invisible language of the world.

Now, a lot of people think mathematics is only about giant scary equations written by sleepy scientists at 3 AM. But real math is mostly about patterns, logic, problem-solving, and figuring things out.

For example:

If an F1 driver completes 1 lap in 90 seconds, and another driver completes it in 88 seconds, who is faster?

Simple.

The second driver.

But now imagine calculating that difference over 70 laps while tracking tire wear, fuel use, weather changes, pit stop timing, and engine temperature ALL at once.

That's why F1 teams need many mathematicians, engineers, strategists, and analysts working together constantly during races.

Some calculations happen in milliseconds.

Literally faster than blinking.

Igloo: "Numbers are predictable. People are not."

Jaskirat: "That sounds deep."

Igloo: "It also describes Igneous perfectly."

And here's something crazy:

Some real F1 engineers were amazing at mathematics when they were students. They studied algebra, geometry, graphs, measurements, statistics, and physics for years before helping build race cars capable of driving over 350 km/h.

So no, mathematics isn't "useless."

Without it:

Bridges wouldn't stand

Rockets wouldn't launch

Planes wouldn't fly

GPS wouldn't work

F1 wouldn't exist

Even game developers use mathematics constantly.

Without math, your favorite games would probably look like spinning potatoes floating through space.

Jaskirat: "Honestly… I'd still play that."

Igloo: "…I know."

And here's the coolest part:

You do NOT need to become some super-genius overnight.

Nobody wakes up one morning and suddenly calculates rocket trajectories while eating cereal.

Mathematics is learned step by step.

Corner by corner.

Lap by lap.

That's exactly how racing works too.

Even champions start small.

So throughout this chapter, we'll explore how mathematics connects to:

1) Speed

2) Distance

3) Time

4) Graphs

5) Percentages

6) Geometry

And yes…

There will also be tons & tons of jokes.

Unfortunately.

Jaskirat: "What do you call a number that can't stay still?"

Igloo: "…What?"

Jaskirat: "A roamin' numeral."

Igloo: "I have left an ancient crystal kingdom for THIS?!"

Speed

Alright future champions, it's finally time to talk about one of the biggest parts of Formula 1…

SPEED!!

Speed is basically how fast something moves from one place to another in a certain amount of time. In simple words:

More distance in less time = more speed.

Sounds simple, right?

Well… in F1, it becomes WAY crazier.

In Formula 1, speed is EVERYTHING. Without speed, F1 cars would just become very expensive traffic.

Teams spend millions trying to make cars even slightly faster because even tiny differences matter. Sometimes races are won by less than one second after nearly two hours of racing. Speed in F1 depends on many things working together:

the engine, aerodynamics, tires, grip, weather, and driver skill. If even ONE thing goes wrong, the whole car becomes slower.

Igloo: "So speed is not just power."

Jaskirat: "Exactly. A fast car without control is basically a shopping cart with confidence."

Now imagine Formula 1 without speed…

Drivers would slowly drive around the track like they're searching for parking at a supermarket. Pit stops would feel longer than movies, and overtakes would take forever. Honestly, it would probably become the world's most expensive traffic jam.

Real F1 cars can go over 350 km/h. That's faster than many airplanes during takeoff. At those speeds, drivers must react instantly while braking, turning, and avoiding crashes at the same time.

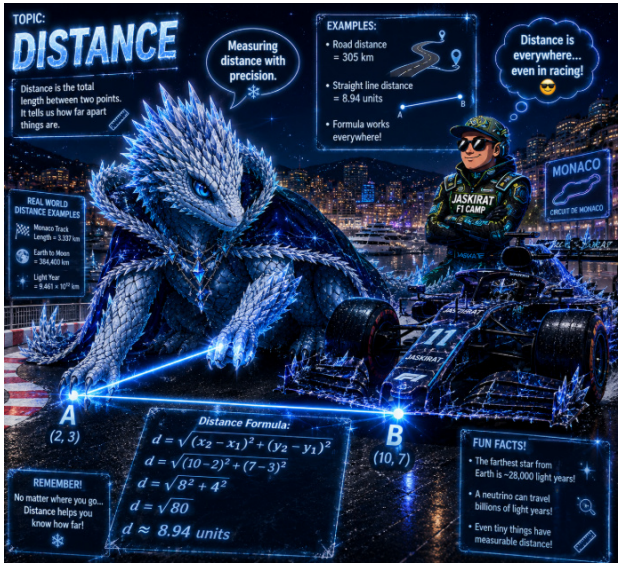
At tracks like Monza in Italy, drivers reach insanely high speeds on long straights. The surroundings blur past while drivers still have to brake perfectly for corners just seconds later. One tiny mistake can ruin the entire race.

That's why speed is not only about going fast.

It's about control, focus, timing, and precision.

Distance

Jaskirat: "Alright, we already learned about speed… but speed alone is useless if you don't know how far you're actually travelling."



Igloo: "Exactly. Distance tells us how much ground something covers while moving. In racing, every metre matters."

Distance is simply how far an object moves from one place to another.

For example, if an F1 car drives around a 5 km track one time, the distance travelled is 5 kilometres. Easy, right?

But in Formula 1, distance becomes super important because races are very long. Drivers don't just go fast — they must calculate fuel, tires, pit stops, and timing depending on how much distance is left in the race.

One wrong calculation and… goodbye victory.

Imagine an F1 driver forgetting the race distance and celebrating halfway through the race.

That would be like finishing page 5 of homework and yelling:

"I'M FREEEEEE!"

…before seeing there are 12 more pages…

Igloo: "Mathematics prevents embarrassment."

Why Distance Matters in F1

Distance helps teams calculate:

Fuel needed

Tyre wear

Pit stop timing

Race strategy

Remaining laps

Without distance calculations, races would become complete chaos. Cars could run out of fuel, tires could explode from overuse, and some drivers might accidentally pit on the final lap for no reason .

Jaskirat: "Honestly… that last one almost happened once."

Igloo: "Now let us solve a simple problem together."

Suppose an F1 car travels 10 kilometres every lap.

If the driver completes 4 laps, what total distance did the car travel?

We simply do:

So the answer is 40 kilometres!

That means:

10 km in Lap 1

10 km in Lap 2

10 km in Lap 3

10 km in Lap 4

Total = 40 km

Igloo: "See? Mathematics is not scary. It is simply organized thinking."

Jaskirat: "Meanwhile Igneous once counted race laps using chicken nuggets."

A real Formula 1 race is usually around 305 kilometres long. That means drivers travel an enormous distance at insane speeds while making split-second decisions the entire time.

To compare:

Driving to school = maybe a few kilometres

An F1 race = hundreds of kilometres at over 300 km/h

And drivers still have to turn perfectly, avoid crashes, protect tires, and overtake opponents while doing all that.

Igloo: "Which is why calculations are as important as speed itself."

Time

Alright future champions… Now let's talk about something SUPER important in Formula 1 — time.

In F1, time is everything. A race can literally be won or lost by 0.001 seconds .

That's faster than a blink, faster than dropping your pencil in class, and probably faster than Igneous running to free food .

Time is basically how long something takes to happen.

How long a lap takes.

How long a pit stop takes.

How long a driver brakes before a corner.

Even tiny bits of time matter in racing because F1 cars move at insane speeds. One small delay can mean another driver zooms past you like a rocket.

Jaskirat: "One bad turn can destroy your whole lap time. Pain!"

Igloo: "Time is an illusion… but its effects are not."

Imagine this:

Two F1 cars are racing toward the finish line.

Car A finishes in 1 minute 32 seconds.

Car B finishes in 1 minute 33 seconds.

That means Car A wins by only ONE second.

Doesn't sound huge… but in F1, one second is MASSIVE.

Without time calculations, Formula 1 would become complete chaos. Drivers wouldn't know when to pit, engineers couldn't plan strategies, and races would basically turn into.

"Uh… just drive and hope for the best."

Even pit stops depend on time. Modern F1 teams can change all four tires in around 2 seconds.

That's faster than most people can sit properly in a chair.

Igloo: "Human reflexes are… surprisingly terrifying."

Here's a simple example:

If a driver completes:

1 lap in 30 seconds

and does 4 laps

Then:

30 × 4 = 120 seconds

120 seconds = 2 minutes

So the total race time would be 2 minutes.

Igloo: "See? Mathematics makes racing easier to understand."

Jaskirat: "Meanwhile Igneous still uses his fingers to count snacks."

Real F1 teams also study lap times every single race. Engineers compare:

fastest laps

average speed

pit stop times

reaction times

tyre performance over time

That's how teams know when to attack, defend, or overtake.

And now… joke Time

Why did the F1 driver bring a clock into the race car?

Because he wanted to improve his "lap TIMES!"

Why was the stopwatch so calm?

Because it knew how to handle pressure.

Why did Igneous fail at timing laps once?

Because he stopped counting after seeing a sandwich

Graphs

Alright future champions, we've now travelled quite far through the world of mathematics. We started with speed, distance, and time—the basic building blocks of racing calculations. But eventually, there comes a point where there are simply too many numbers to look at one by one. Imagine an F1 engineer trying to study thousands of lap times, tyre temperatures, fuel levels, weather readings, and speed measurements all at once. After about five minutes, their eyes would probably start doing donuts around the garage.

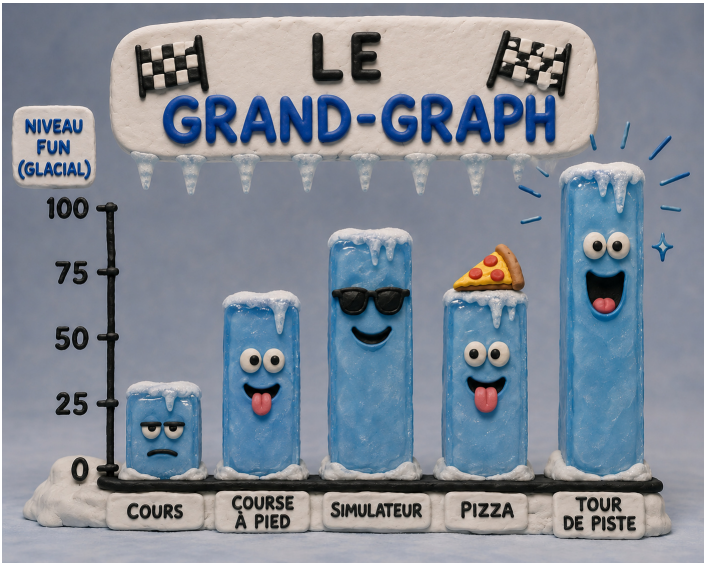
Here's another:

"Lap 1 was 92.4 seconds, Lap 2 was 91.8 seconds, Lap 3 was 91.2 seconds, Lap 4 was 90.9 seconds..."

After about ten laps, everyone's brain would look like a tangled pit-lane radio cable.

This is where **graphs** enter the race.

For some reason this image is in French :P. →



A graph is a visual way of showing information. Instead of reading hundreds of numbers, a graph allows us to spot patterns, trends, and problems almost instantly. Think of it as turning a giant wall of calculations into a picture your brain can understand much faster.

Jaskirat: "Graphs are basically the reason engineers still have their sanity."

Igloo: "Most of it, anyway."

Graphs are used everywhere in Formula 1. Teams monitor lap times, tyre wear, fuel consumption, battery levels, weather conditions, brake temperatures, and even the amount of grip available on different parts of the track. During a race weekend, teams collect millions of pieces of information. Without graphs, understanding all that data would be nearly impossible.

Now let's look at something that might seem difficult at first.

Imagine an engineer creates a graph showing tyre grip during a 30-lap race. The graph stays almost flat for the first 18 laps, meaning the tires are performing well and giving the driver strong grip. However, after Lap 18, the line suddenly begins dropping much faster. By Lap 24, the graph is falling steeply.

At first glance, it just looks like a line moving downward.

But what is the graph actually telling us?

The tires are wearing out faster than expected. The driver may still feel fast, but the graph is warning the engineers that performance is about to drop dramatically. If they wait too long, valuable seconds could be lost every lap. If they react quickly, they can bring the driver into the pits before the problem becomes serious.

That's the amazing thing about graphs. They don't just tell us what has happened—they often help us predict what is going to happen next.

Quick Challenge

Suppose you see a speed graph that rises from 120 km/h to 320 km/h, suddenly drops to 90 km/h, and then quickly rises again.

What do you think happened?

Take a guess before reading on!

Most likely, the driver accelerated down a straight, braked heavily for a corner, then accelerated out of the turn.

See? The graph is actually telling a story.

Igloo: "Every graph has a story hidden inside it."

Jaskirat: "And sometimes that story is 'someone made a terrible strategy decision.'"

Graphs aren't only used in racing. Scientists use them to track climate change , doctors use them to monitor heart rates , businesses use them to track profits , and even schools use them to measure student performance .

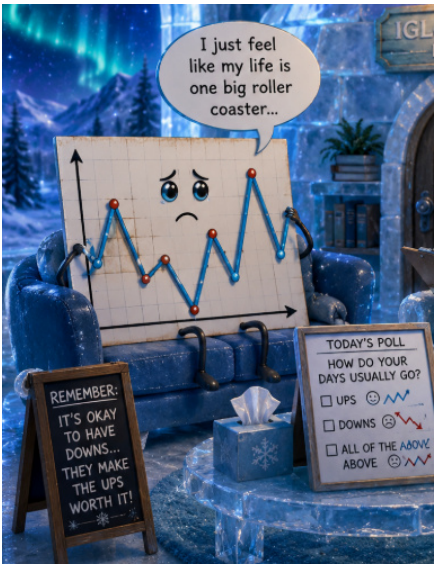
So why have we moved from simple maths to graphs?

Because simple maths helps us calculate answers.

Graphs help us understand entire situations.

It's like the difference between looking at one puzzle piece and seeing the whole puzzle.

Joke Time:



Why did the graph go to therapy?

Because it had too many ups and downs!

Why do engineers love graphs?

Because graphs don't argue over the radio.

What did the graph say to the calculator?

"Stop giving numbers and show me the bigger picture!"

Jaskirat: "How many engineers does it take to read a graph?"

Igloo: "One…"

Jaskirat: "Then why are there twenty standing around it?"

Igloo: "To make it look important! -_-"

& these mini Igloo dragonets make it 10x funnier!! →

Geometry

Alright campers, we've now explored speed, distance, time, and graphs. But now we're entering one of the most important areas of mathematics in engineering, racing, architecture, game design, and even nature itself.

Welcome to **Geometry**.

At first glance, geometry might look like a bunch of shapes, angles, lines, circles, and triangles sitting around doing absolutely nothing. But don't let it fool you.

Geometry is one of the reasons Formula 1 cars can corner at incredible speeds, airplanes can fly, bridges can stand, and buildings don't randomly collapse onto the ground.

Geometry is the branch of mathematics that studies shapes, sizes, angles, positions, and how different objects fit together in space. In simple words, geometry helps us understand the shape of the world around us.

Igloo: "Everything has geometry."

Jaskirat: "Even my race car."

Igloo: "Especially your race car."

Take a look around you right now. Your book is probably a rectangle. A clock might be a circle. A slice of pizza is usually a triangle. Even an F1 track is made from curves, angles, and carefully designed shapes. Geometry is literally everywhere.

But why does Formula 1 care so much about geometry?

Because tiny shape changes can completely change a car's performance. Engineers spend thousands of hours designing wings, suspension parts, tires, and bodywork. Every curve and angle has a purpose. A wing tilted by only a few degrees can affect airflow, grip, and speed. That's why geometry isn't just school mathematics—it's one of the most powerful engineering tools in the world.

Now let's look at a more advanced example.

Imagine an engineer is designing a corner for a new race track. The corner is shaped like part of a circle. If the curve is too tight, drivers must slow down too much. If it's too wide, the corner becomes too easy and less exciting.

The engineer uses geometry to calculate the radius of the corner and determine how fast drivers can safely travel through it. Without geometry, track designers would basically be drawing random squiggles and hoping for the best.

Jaskirat: "That's probably not the safest way to build a race track."

Igloo: "It is also an excellent way to lose your engineering license."

Geometry is also used inside the cars themselves. The suspension angles affect how tires touch the road. The front wing uses carefully designed shapes to guide air around the car.

Even the driver's seat position is calculated using geometry to maximise comfort, visibility, and safety.

In fact, modern Formula 1 teams use so much geometry that entire departments exist just to study shapes, angles, and airflow patterns.

Geometry Challenge: The Wing Mystery

You're an F1 engineer preparing Jaskirat's Neon Raptor Vortex GT-X for a championship race.

During practice, the team notices the car is sliding a little in fast corners. One engineer has an idea:

"What if we tilt the front wing upward by just 2°?"

Everyone looks at the wing.

The change is tiny, actually nothing.

So tiny that most fans in the grandstands probably wouldn't even notice it.

What do YOU think?

Would changing the wing by only 2 degrees:

A) Do almost nothing

B) Make the car slightly different

C) Significantly affect the car's performance

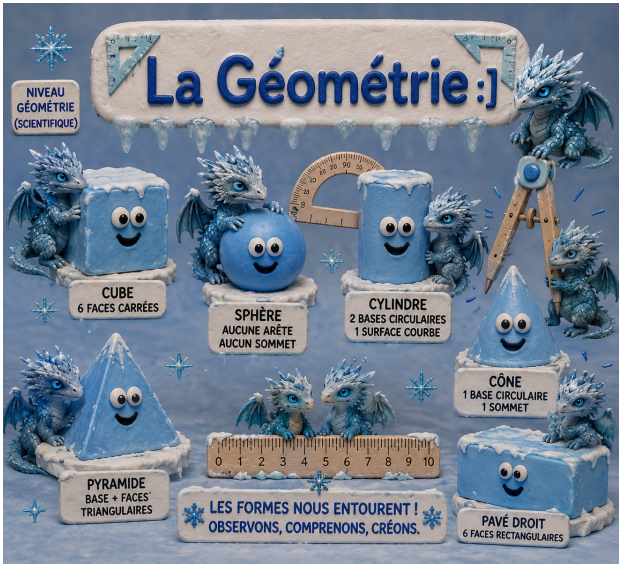
Take a guess before reading the answer!

↓

↓

it 's (c) *_*!!

Even a tiny angle change can affect how air flows around the car. More air pushing downward can create extra downforce, helping the car grip the track better in corners. However, it can also create more drag, making the car slower on straights.



That means a small 2° adjustment could affect:

1. Cornering speed
2. Top speed
3. Tyre wear
4. Fuel consumption
5. Overall race performance

In Formula 1, engineers often spend hours testing changes that look almost invisible. Sometimes a tiny

angle adjustment can be worth several tenths of a second per lap—which can be the difference between standing on the podium or watching someone else lift the trophy!

Igloo: "This is why geometry matters."

Jaskirat: "Imagine losing a race because of 2 little degrees!!"

That's the power of geometry.

It teaches us that small changes in shape can create huge changes in results.

And that's not just true in racing. Geometry helps architects design skyscrapers, game developers build worlds, scientists explore space, and engineers create everything from roller coasters to spacecraft.

Igloo: "Geometry is where mathematics begins taking shape."

Jaskirat: "I see what you did there."

Joke Shape-Shift!

Why was the triangle so confident?

Because it always had a point!

Why did the circle get promoted?

Because it was well-rounded!

What did the angle say after winning the race?

"That was the right direction!"

Jaskirat: "Why don't geometry teachers ever get lost?"

Igloo: "Why?"

Jaskirat: "Because they always know all the angles!"

Igloo: "...Acceptable. -_-"

Conclusion – THE GRAND FINISH!!

Well, campers... we've reached the finish line of this chapter.

When Igloo first arrived at Formula Racing Camp, she came to help us solve one problem: a certain judge who somehow believed Crimson Cow could win races even when they crossed the finish line second.

But along the way, we discovered something much bigger.

Mathematics isn't just about solving questions in a textbook. It's one of the most powerful tools in the world. We saw how speed, distance, time, graphs, and geometry help drivers, engineers, scientists, architects,& our beloved game developers solve problems, big or small every day.

In Formula 1, mathematics helps teams predict strategies, understand tyre wear, improve aerodynamics, analyse data, and build faster cars. Without mathematics, modern F1 would be impossible. There would be no race strategies, no lap-time analysis, no engineering breakthroughs, and probably a lot more cars accidentally ending up in gravel traps.

Igloo: "Mathematics is not about memorising numbers. It is about understanding the world."

Jaskirat: "And occasionally proving that Formula Racing Camp actually won the race."

From simple calculations to advanced graphs, every topic we explored was another tool added to your toolbox. You might not become an F1 engineer tomorrow, but every great engineer, scientist, inventor, and strategist started by learning the basics first.

And remember: mathematics isn't about being fast.

It's about thinking.

Sometimes the smartest solution isn't the most complicated one—it's the one that helps you understand the problem clearly.

Suddenly, a loud voice echoed from down the hallway...

Igneous: "WAIT... THE CHAPTER'S OVER?!"

Jaskirat: "...He missed it?!?!"

Igloo: "Unfortunately."

Igneous came running into the room looking completely shocked.

Igneous: "I MISSED THE ENTIRE CHAPTER?!"

Jaskirat: "You slept through all of it bro!!"

Igneous: "I MISSED SPEED?!"

Jaskirat: "Yep."

Igneous: "DISTANCE?!"

(Don't ask why this image is supa goofy!) →

Igloo: "Also yes."

Igneous: "GRAPHS?!"

Jaskirat: "Those too."

Igneous: "NOOOOOOOO!!!!! THAT'S NOT N'ICE!!!"

Igloo: "To be fair, you were asleep for approximately eight hours and forty-three minutes."

Igneous: "YOU TIMED IT??!!!"

Igloo: "Of course."

Jaskirat: "That's actually one of her hobbies."

Igneous sat down dramatically.

Igneous: "Fine. I'll read the chapter later."

Jaskirat: "Good idea."

Igneous: "Can you give me a summary 19 pages long?"

Igloo: "No."

Igneous: "...Worth a try!!"

The End of Book-1... The Start of Something Bigger!!

And with that, future champions, we have reached the end of the very first Formula Racing Camp book.

Over the course of this adventure, we've explored the incredible world of Formula 1—from the science of aerodynamics and the power of engines to the mathematics that helps teams make split-second decisions. We've learned how speed, distance, time, graphs, and geometry help shape every race. More importantly, we've discovered that learning isn't just about memorising facts—it's about understanding how the world works and using that knowledge to solve problems.

Along the way, we've met amazing teammates, survived questionable experiments, witnessed Igneous doing... Well, Igneous things , welcomed Igloo to Formula Racing Camp, and proved that curiosity can be just as powerful as horsepower.

But every great race eventually reaches the checkered flag. The good news is that every finish line is also the start of a new journey. The lessons you've learned in this book are only the beginning. There are still new technologies to discover, new races to win, new mysteries to solve, and plenty of adventures waiting for the campers of Formula Racing Camp.

Whether you dream of becoming a racing driver, engineer, inventor, scientist, designer, programmer, or something completely different, remember this:

Great champions are not born knowing everything.
They become great because they never stop learning.

So keep asking questions.
Keep exploring.
Keep creating.
And never be afraid to try something new!

Because while this may be the end of– Formula Racing Camp: Season 1...

...but the story of Formula Racing Camp is only just getting started!

See you in the next adventure, future champions!!!

FUN FACTS

Hello again, campers! Worried that the book ended before you even found out who's the best F1 Driver? Don't worry, here are some cool facts you can know about in F1!

• The driver with the most WDCs - It's actually a tie between Lewis Hamilton and Michael Schumacher, who each hold 7 titles.

• The constructor with most WCCs - Scuderia Ferrari, with 16 titles.

• The First F1 Race - May 13, 1950 at Silverstone Circuit.

• The Main First F1 Teams - Alfa Romeo, Scuderia Ferrari, Maserati, and ERA (English Racing Automobiles).

• Most Race Wins - Lewis Hamilton at 106.

• Most Podiums - Lewis Hamilton at 207.

• Most Wins in a Season - Max Verstappen at 19 wins in 2023 Season.

• Most Points in a Season: Max Verstappen at 575 points in 2023 Season.

• Most Consecutive Race Wins: Max Verstappen at 10 wins in a row in 2023 season.

• Record for most records for all-time F1 History - Lewis Hamilton

• Records for most record for in-season F1 History - Max Verstappen

Campers, you have officially reached the end of Formula Racing Camp Season 1, but again, don't worry! We have more exclusive Formula Racing Camp content in our website https://formularacingcamp.com and our YouTube channel https://youtube.com/@wheref1rulestheworld . We also have puzzles at https://formularacingcamp.com/puzzles

Stay tuned for Season 2!! ;)

{The image may have some distortions :P}

Igloo: "N'Ice memories never melt—even if a few distorted pixels do! Plus we have said the phrase "N'Ice" 3 TIMES; now 4!!!"

IMAGE ATTRIBUTIONS

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